Displayed Price, Hidden Clearing: Fallback and Selection in AI Inference Markets

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Abstract

Open-weight models create a real-time market for partially substitutable inference: harnesses originate token demand, routers choose where to buy execution, and providers sell capacity under common model labels. We separate the displayed price menu from the hidden clearing process using five-minute quotes, public enforcement state, cross-router catalogs, and randomized owned requests. Prices are sticky and often common across routers while eligibility and enforcement move at unchanged prices. We introduce a randomized decomposition of fallback and delegated selection, with outcomes released only at its frozen gate. The evidence supports a two-layer procurement market, not a public order book; it does not yet identify market-wide welfare or provider intent.

1 Introduction

Open weights separate a model's production technology from the firm that executes it. The traded object is a completion of a declared model on a particular task. It is partly commodity-like—multiple providers can run the same weights—but not perfectly fungible: sampling, tool calls, serving stacks, latency, and hidden capacity can change whether and how the task completes.

The resulting market has a distinct vertical stack. An end user delegates to a harness or agent runtime, such as Hermes, which can transform the task, choose a model, call tools, set a budget, and retry. A router such as OpenRouter aggregates purchase requests across harnesses and privately dispatches each request to an eligible inference provider. Providers supply GPU-backed, perishable service capacity under a common model label. The model is therefore a compatibility standard and production technology; the harness is a demand-side broker and order-flow originator; the router is a two-sided dispatcher and procurement platform; and the provider is a capacity-constrained service seller.

This organization creates a prior measurement question: when providers display a price for the same model, which part of execution is cleared by that price and which part by hidden eligibility, provider scoring, and fallback? Welfare and conduct questions come afterward. Without observing the clearing margin, a cheap quote can be mistaken for firm supply and a high aggregate fill rate can be mistaken for provider reliability. The answer cannot come from token prices alone because the router, rather than the public quote surface, clears eligibility and quantity.

That question is difficult to answer from posted prices alone. The price surface is public, but the allocation rule and capacity state are private. The same cheap quote can be visible to every user while being unavailable to a particular request. Conversely, a router can deliver a high fill rate even if no individual quote is firm, by substituting across providers. Treating the public surface as an order book therefore confuses displayed liquidity with delivered liquidity. Treating the router as a decentralized-exchange aggregator makes a related mistake: an inference request is not atomically split across providers, the price is not an executable commitment, and selection depends on hidden eligibility and quality.

We characterize the market with a measurement design that joins four kinds of evidence: high-frequency quote surfaces, router-side operational aggregates, historical price records, and our own randomized micro-purchases. The design keeps three estimands separate. Public observations measure the displayed menu; published operational aggregates measure congestion and enforcement; and owned probes measure the realized policy for one account and workload. None alone identifies market-wide flow, private capacity, or provider cost.

The paper asks one narrow question: how much of market clearing occurs through the displayed price and how much occurs through hidden eligibility, provider selection, and fallback? We organize the answer around a two-layer market. The display layer contains public provider–model menus. The clearing layer maps a request into an eligible provider and, if necessary, a sequence of fallback attempts. This organization replaces an earlier accretive version of the paper that treated price atoms, revenue elasticities, entry, and steering as coequal results. Those analyses remain useful diagnostics or companion results, but they are not evidence for the focal clearing claim.

1. **A market and measurement framework.** Open-weight execution is a procurement market for a partially substitutable service. The model is a compatibility standard; providers sell perishable execution; the router is a dispatcher and delegated buyer; and the harness is a demand-side broker. We define separate observables for displayed menus, public enforcement state, simulated allocation, owned execution, and private router logs. This hierarchy makes clear why public simulated "routed share" is not realized demand.
2. **Displayed prices do not exhaust the clearing state.** The public menu changes slowly while router-published rate-limit and derank state changes frequently at unchanged price. The H82 event paths are descriptive because their pretrends fail. The separately frozen H84 test rejects one natural stale-cheap prediction but has a nonzero backward placebo. A first simultaneous cross-router catalog has 28 of 29 exact provider–model price matches, with Wilson interval $[0.828, 0.994]$; because there is only one cross section, this is coverage evidence rather than a pass-through estimate.
3. **A randomized decomposition matched to the mechanism.** In H81 we hold the displayed cheapest provider fixed and randomize among cheapest-only execution, explicit price order with fallback, and full delegation. The two primary contrasts isolate the option value of fallback and the additional value of hidden selection for the owned account. Assignment replay and policy compliance are outcome-free. At the pinned data revision the arm counts are 32, 24, and 28, below the original 40-per-arm release gate; no H81 outcome is reported. The eligible support currently contains only two repeated models, which limits transport even after release. We therefore froze H95 before its first request: an independent fixed-horizon replication over 120 three-model triplets (360 first-position assignments, exactly 120 per arm) drawn from the broader rank-7–30 eligible frontier. H81 and H95 are never pooled, and no H95 outcome is queried before all 120 plans exist. H95's prerelease analyzer uses exact nuisance-conditioned pairwise Fisher inference, simultaneous bounded- outcome design intervals, explicit measurement-missing bounds, row-level treatment-control coverage, and registered time and whole-triplet transport diagnostics; it also exposes the sequential-position interference assumption rather than treating it as automatic. A separately frozen position-zero estimator uses the first model block only to remove carryover from earlier H95 blocks, at a disclosed loss of precision.
4. **Identification and design.** We prove the accounting decomposition and, after removing the gate-hitting terminal block, the conditional design-unbiasedness of the fixed-count arm means (equivalently, conditional Horvitz–Thompson contrasts). The frozen release reports exact pairwise-conditional Fisher randomization tests with Holm adjustment, descriptive Newcombe intervals,

and a conservative finite-population confidence set simultaneous over the three policy means. Unknown or malformed outcomes are not failures: they suppress point inference and enter $[0, 1]$ worst-case bounds together with missing or noncompliant treatment records. We also characterize what coarsened public timing cannot identify. The design implication is execution-contingent procurement—prices, eligibility, fallback, and reliability must be evaluated jointly—not a claim that the current data identify a welfare optimum, collusion, or literal front-running.

These facts jointly imply a sharper market analogy. The model is a product standard or protocol; an inference provider is a dealer with a revocable quote and hidden inventory; the router is a dispatcher and delegated purchasing agent; and the harness is an application-side broker controlling retries and fallback. The closest mature markets are request-for-quote dealer markets and platform procurement, not constant-function automated market makers. The analogy matters because market quality depends on admission, substitution, and steering rules, not only on quoted spreads.

Our results contribute to three literatures. Work on language-model routing optimizes quality-cost choice across models Chen et al. [2023], Ong et al. [2024], Hu et al. [2024]; we instead measure competition and execution among providers of the same model. Brown and MacKay [2023] show how heterogeneous pricing technology changes competition; we provide the first high-frequency pricing-technology screen for inference providers, while making its short-panel limits explicit. Platform design work studies prominence and algorithmic pricing [Johnson et al., 2023]; our audit turns its comparative static into a statistic that a key-holder can reproduce from realized selections.

2 Institutional setting and economic objects

2.1 The transaction

A request names a model m, supplies an input, and may specify provider preferences. For provider i at time t, the displayed menu contains an input token price p^{in}_{imt} and an output token price p^{out}_{imt}. For a fixed workload with a input and b output tokens, the displayed money quote is

$$q_{imt}(a, b) = ap^{\text{in}}_{imt} + bp^{\text{out}}_{imt}. \tag{1}$$

The platform can additionally observe health, throughput, latency, context-window compatibility, account-specific limits, and contractual eligibility. Let $e_{imtu} \in \{0, 1\}$ denote eligibility for user u, and let c_{imt} denote unobserved deliverable capacity. The default router selects a provider according to an unknown score

$$i^*_{mtu} = \arg\max_i S(q_{imt}, e_{imtu}, c_{imt}, z_{imt}, h_u), \tag{2}$$

where z collects measured quality and h_u is account state. A pinned request sets the candidate set to one provider and can disable fallback. Comparing default and pinned execution therefore measures the value of delegation, not merely a price effect.

The request is served by one provider. Fallback may create multiple attempts, but this is sequential substitution rather than splitting one prompt across a portfolio. For the interfaces studied here, the route cannot generally be observed as a firm executable allocation before purchase. This absence of a public pre-trade route is why literal order-flow front-running is not identified from public data.

2.2 Measurement levels

We use an explicit observability hierarchy. It prevents a common error in this market: assigning the identifying content of realized flow to a price-only simulation.

Level	Object	Identified use
L0	Public quote surface	Price levels, gaps, ties, grids, and repricing events
L1	Documented policy simulation	Mechanical ranking or allocation proxy only
L2	Public operational aggregates	Congestion, rate-limit, derank, and enforcement state
L3	Owned realized requests	Selected provider, success, latency, cost, and fallback for the probed account
L4	Provider or router logs	Capacity commitments, full attempt order, and cross-user flow

Table 1: Observability hierarchy. The core paper uses L0–L3. No claim that requires cross-user order flow or private provider cost is made.

Public token-allocation displays sometimes expose a deterministic price-weighting proxy. When a documented inverse-square proxy is available, we compute

$$\widehat{s}_{imt} = \frac{q_{imt}^{-2}}{\sum_j q_{jmt}^{-2}}, \qquad \frac{\partial \log \widehat{s}_{imt}}{\partial \log q_{imt}} = -2(1 - \widehat{s}_{imt}). \tag{3}$$

Equation (3) is a policy arithmetic identity. It is not demand elasticity and is never substituted for realized routing in our causal claims.

3 Data and research design

3.1 Capture system

The live panel snapshots public provider-model menus every five minutes. Each record stores a UTC capture time, stable provider and model identifiers, input and output token prices, capacity ceilings when displayed, and router-published five- and thirty-minute enforcement aggregates. Captures are append-only and mirrored to remote object storage. The analysis also uses a three-year historical quote registry for lifecycle checks, public model-level token-allocation records, and owned generation records for requests made with our study key.

The paper release distinguishes the current acquisition cut from older frozen discovery vintages. At immutable public-input revision `b389923ad771`, the endpoint panel contains 2,004,680 distinct provider–model listing observations from 2,116 capture runs on 11 UTC dates, 2026-07-07 through 2026-07-17. The corresponding exact-distinct raw-record count is 2,013,866; the difference preserves 9,186 same-listing raw variants, all due to availability state rather than price or capability conflicts. The outcome-free randomized-design status is separately pinned at revision `30b430e2a09`: H81 contains 84 verified first-position blocks, with arm counts 32, 24, and 28 and exact assignment replay. Its earlier eligibility audit contains 74 candidate rows across 37 runs; H80 remains a separate study with 156 verified first-position blocks over 40.7 hours. The public cross-router collector has two snapshots per router over 0.054 days. Its latest catalog has 107 Glama rows, 114 NemoRouter rows, and 563 Requesty rows, but no price event.

H94 was activated as a future-only longitudinal protocol at 04:30:20 UTC. Its 03:30 discovery cross-section is ineligible; one later prospective snapshot per router is present, with zero price transitions or matched common shocks. H95 was frozen in commit `00351dd` at 04:44:10 UTC before its first request. Five written triplets are now present: all 15 first-position assignments replay and comply, they cover nine unique models (effective model count 7.76), and no outcome field was queried. The newest triplet's three provider-control records are auditable and all pass; the preceding 12 remain legacy-unverified. Its confirmatory horizon remains the first 120 written triplet plans, not a balance-based stopping time.

The price-atom, repricing-hazard, steering, and Brown–MacKay analyses below were frozen on earlier nine-day discovery cuts. They are retained to document the falsification path and are not silently refreshed into confirmatory evidence. Local staging outputs that are not rebuilt from the pinned revision are excluded from current-status claims.

Panel	Rows / attempts	Span	Primary role
Endpoint acquisition cut	2,004,680 listings	11 UTC dates	Current prices and public enforcement state
Frozen pricing discovery cut	1,747,960 listings / 313 changes	9 days	Price atoms, hazards, and cadence screens
H81 first-position assignments	84	ongoing	Fallback/selection release-gate support
H81 candidate eligibility	74	37 runs	External-support and exclusion funnel
H80 first-position assignments	156	40.7 hours	Separate four-arm replication support
Cross-router public catalogs	2 snapshots/router	0.054 days	Coverage and exact price equality only
H95 prospective replication	5 triplets / 15 blocks (12 legacy-control rows)	120-triplet fixed horizon	Independent multi-model fallback/selection replication
Historical quote registry	multi-year	3 years	Lifecycle and duration sensitivity

Table 2: Paper evidence panels at the pinned revision. Earlier discovery cuts remain frozen; outcome-gated studies expose assignment and support counts only.

3.2 Cleaning and event construction

Prices are converted to dollars per million tokens before comparisons. We use standard provider variants and deduplicate exact source records by capture time, model, provider, tag, endpoint fingerprint, and raw record. Distinct raw variants at the same listing and time are retained until price agreement is checked; a price or capability conflict fails closed. We then collapse price-identical variants for listing-level analysis. A price event is the first observation at which either token price differs from the preceding observation for the same provider-model pair. Daily events are rebuilt chronologically from the strictly prior daily state and replace, rather than merge with, the old event partition. At the pinned revision, all 3,219 derived field-level events through July 16 exactly match a clean chronological rebuild. Five-minute timing creates interval censoring within a capture bin; we do not infer within-bin order. The first price spell in a panel is left-censored and is excluded from moments that require age.

A pre-publication integrity audit found that earlier daily recompactions could fold an old day against a later global state and then merge stale events. It also found 1,532,963 exact duplicate physical rows in the pre-repair endpoint bundles. The repaired revision removes those overlaps from every completed date and rebuilds the event ledger. The still-open July 17 buffer contains 256 physical duplicates; listing-level analyses deduplicate them and completed-day event claims stop on July 16. Earlier draft statistics that do not reproduce from this ledger are superseded rather than grandfathered.

Operational counters are router-published aggregates, not raw request logs. Their reporting can be intermittent. Regressions that use utilization or rate-limit state include a missingness indicator and are repeated on the complete-reporting subsample. Status-derived incidents are labeled as heuristics unless the source publishes exact incident timestamps. Owned probes record only metadata needed for the study–assignment, policy, provider, latency, token count, cost, and status–and never retain prompt payloads.

3.3 Claim discipline

The empirical sequence was adaptive, so we distinguish discovery, protocol, amendment, and confirmation. Early descriptive work generated the price-atom, firmness, and enforcement hypotheses. The fixed-order probe confounded policy with position and is retained only as a design failure. H81 was written before its first request and has an original 40-per-arm gate, although a legacy analysis inadvertently exposed aggregates from its first two blocks. H80 is a separate experiment: its original gate was also 40 per arm, and its later 500-per-arm promotion threshold is disclosed as a post-outcome, pre-original-gate amendment. A proof audit further corrected H81's stopped-prefix inference before confirmatory release by excluding the gate-hitting block and conditioning on preterminal arm counts. H82 is a retrospectively frozen descriptive event study and H84 is a retrospectively preregistered discovery falsification. H94 is a future-only protocol whose activation cutoff excludes the discovery cross-section. H95 is an independent prospective fixed-horizon replication frozen before its first request; it cannot inherit or pool H81 outcomes. Every change is listed in the public amendment ledger. A test below its frozen support is called power-gated, never a null result.

4 Fact 1: prices are administered menus

4.1 Frequency, size, and the clock

The corrected frozen nine-day panel contains 313 nonzero completion-price changes, or 0.04257 changes per provider–model day. The inverse-frequency duration is 23.5 days. Conditional on a change, the median absolute log change is 0.095 and 70.3% are cuts. These are incidence and event-size diagnostics, not estimates of a continuous-time structural hazard.

Raw pooled price-change kurtosis is 215.27 because providers and model scales differ. After standardizing within provider for providers with at least three events, kurtosis is 6.23 over 307 changes, numerically close to the Calvo benchmark of six discussed by Alvarez et al. [2016]. This corrected moment does not support the earlier draft's "between menu cost and Calvo" reading. We use it only as a sufficient-statistic diagnostic: the short panel leaves many providers with too few changes for stable within-unit moments, provider-level standardization is coarser than endpoint-level standardization, and observation costs can mimic infrequent adjustment Alvarez et al. [2011].

4.2 A nested repricing hazard

For provider-model pair k on day d, let $Y_{kd} = 1$ if a price changes. We fit a nested logit ladder

$$\mathbb{P}(Y_{kd} = 1 \mid X_{kd}) = \Lambda(\alpha + T_{kd} + G_{kd} + C_{kd} + R_{kd}), \tag{4}$$

where T contains age and calendar terms; G contains the absolute and signed gap from the cheapest rival plus GPU-cost changes; C contains published utilization and rate-limit state; and R contains rival raises and cuts in the prior day. The ladder adds these blocks sequentially. Provider activity is included as a frailty proxy, but with only nine days we do not estimate a full unit-frailty duration model.

The corrected frozen ladder contains 7,352 pair-days and 112 events. Duration and calendar terms improve the constant-hazard likelihood ($p = 3.45 \times 10^{-67}$). Adding price gaps and GPU changes improves in-sample fit ($p = 2.04 \times 10^{-6}$); adding congestion does not ($p = 0.419$); adding rival moves again improves in-sample fit ($p = 8.03 \times 10^{-5}$). The checked-in analyzer produces in-sample AUC only. We therefore retract the earlier draft's day-split AUCs and make no claim that any rung survives temporal validation.

The confirmatory temporal test is now executable and frozen before its holdout exists. It excludes the open UTC date, waits for 30 completed quote dates, fits on dates 1–15, and evaluates once on dates 16–30. The event-day outcome is predicted only from prior-close price gaps, GPU changes, congestion, and rival moves; the provider-activity control is estimated from the training prefix only. All rungs use training-standardized ridge logistic regression with $C = 1$, fixed before the holdout and never tuned on it; this replaces an unpenalized fit that could separate. The primary L3 rung has 17 parameters including its intercept. Promotion requires at least ten training events and ten nonevents per parameter, 50 test events and nonevents, ten train/test event dates, and ten test models; a failure yields an insufficient-support verdict rather than a weaker model. The primary contrast is the date-weighted paired Bernoulli log-loss improvement of the lagged-state rung over the duration/calendar rung. Four adjacent-rung one-sided sign-flip tests are Holm-adjusted, with date- and model-cluster bootstrap intervals and leave-one-model-out sensitivity. At revision `4fd167d674f6`, 11 dates are represented but only 10 are complete; the open July 17 date is excluded, 20 completed dates remain, and the program does not query a pricing outcome before the gate. These estimator and support rules were frozen in commit `1719ade` while the holdout remained inaccessible.

Hazard rung	Added state	Incremental evidence	In-sample AUC
Constant	none	baseline	0.500
Time-dependent	duration, clock	LR $p = 3.45 \times 10^{-67}$	0.876
State-dependent	own-rival gap, GPU change	LR $p = 2.04 \times 10^{-6}$	0.879
Congestion	utilization, rate limits	LR $p = 0.419$	0.882
Strategic	rival moves	LR $p = 8.03 \times 10^{-5}$	0.884

Table 3: Corrected frozen nine-day nested repricing hazard. Every AUC and likelihood-ratio comparison is in-sample; the 30-day vintage and temporal validation remain closed.

The appropriate conclusion is narrower than "providers use pricing algorithms." In this short panel, repricing is infrequent and its in-sample hazard is associated with duration, calendar state, and distance from a competitive reference price. We call the quotes administered menus because the money price is revised intermittently while operational state clears elsewhere. Whether the associations generalize, or whether rival terms reflect common costs, algorithmic competition, or coordinated behavior, remains unresolved.

5 Fact 2: price atoms are real, but their mechanism is not identified

Displayed minimum prices cluster on exact common levels. In the frozen panel, 45.9% of 1,311 multi-provider model-days have a tie at the minimum, versus 13.4% under an independent cent-grid null that preserves market thickness, price scale, and empirical provider heterogeneity. A coarser dime-grid null raises the benchmark to 27.6%, still well below the observed atom. This is evidence of a shared discrete menu, not evidence of a particular coordinating agent.

The identity tests remove that stronger interpretation. Conditional on each market's realized price multiset, author-operated endpoints occupy a shared price in 54.3% of eligible models, versus a 53.5% random-endpoint expectation (exact upper-tail $p = 0.466$). Similarly, 20.4% of isolated revisions land on a rival's strictly prior quote, but the matched common-menu benchmark is 13.4%; the 7.0-point excess has model-cluster interval $[-23.1, 12.2]$ points and is dominated by one model. A preregistered same-provider/across-model control and own-menu-novel restriction do not satisfy the frozen promotion rule.

Question	Estimate	Identifying benchmark
Is there a minimum-price atom?	45.9% ties	13.4% independent cent-grid null
Is author identity special?	54.3% shared	53.5% random-endpoint expectation
Do revisions land on rival prices?	20.4% exact	13.4% matched-menu probability
Is strategic following identified?	No	sharp share $[0, 20.4\%]$

Table 4: Price-atom evidence and claim boundary. The atom survives grid-aware nulls; author identity and rival-triggered response do not survive their harder nulls. Full algorithms, intervals, power simulations, and robustness checks are in section A.

Proposition 2 formalizes the boundary: with coarsened snapshot timing and latent public menu state, the same observed exact landings can be generated by scheduled refreshes or rival-triggered revisions. The named-rival response share is therefore sharply bounded only between zero and the observed exact-landing share. The registered known-clock simulation controls false promotion but fails all four empirical calibration diagnostics, so it remains a stress test rather than a fitted mechanism. The full theorem and constructive proof, the detection-threshold proposition, cluster enumeration, and the frozen 15/15 held-out matched-market protocol are reported in the appendix. The main paper uses this fact only to motivate hidden clearing: displayed price equality does not imply equivalent execution contracts or reveal why a quote moved.

6 Hidden clearing: fallback and delegated selection

6.1 Why public operations are not realized routing

Displayed price and operational state move on different clocks. Public rate-limit and derank transitions occur while provider–model token prices remain unchanged. H82 aligns endpoint-specific enforcement onsets with neighboring providers for the same model. The event paths show a broad operational response, but all three frozen pretrend checks fail; we therefore treat the result as a description of the hidden clearing margin, not a causal effect of enforcement. H84 then asks whether a relatively stale cheap quote predicts the next high-capacity event. Its directional prediction is rejected, while a backward placebo is nonzero. This falsifies one simple pickoff mechanism without ruling out adverse selection more generally.

These public events do not reveal which provider served a request. They motivate an owned-traffic experiment in which the displayed provider list is held fixed and only fallback and delegation change.

6.2 H81 treatment decomposition

For each eligible model–time block, the collector obtains one public provider list and uniformly randomizes which of three policies is executed first:

1. N (no fallback): request only the displayed cheapest provider;
2. F (fallback): submit the displayed providers in price order and permit fallback; and
3. D (delegation): submit no provider restriction and permit the router's default selection and fallback.

Only the first request is confirmatory, so arbitrary interference from that request to later attempts in the block cannot bias the focal estimand. The fallback contrast $F - N$ holds the first displayed provider fixed. The hidden- selection contrast $D - F$ holds fallback permission fixed and changes who chooses the order and eligible set. Neither contrast reveals the router's private score.

The release rule is a stopping time: let T be the first block at which every arm has at least 40 first-position assignments. The gate-hitting block has an assignment mechanically related to the preceding imbalance, so it is excluded from estimation. Set $B = T - 1$, let n_p be the realized count of policy p among the preterminal blocks, and let $\pi_p = n_p/B$. For bounded potential outcome $Y_b(p)$, define

$$\widehat{\mu}_p^{HT} = \frac{1}{B}\sum_{b=1}^{B}\frac{\mathbf{1}\{Z_b = p\}Y_b^{obs}}{\pi_p} = \frac{1}{n_p}\sum_{b:Z_b=p} Y_b^{obs}. \tag{5}$$

Proposition 1 (Randomized fallback and hidden-selection decomposition)**.** *Suppose the production policy permutation is independent of the fixed potential-outcome schedule. Conditional on T, the terminal policy, and the preterminal count vector (n_N, n_F, n_D), the assignments of the first $B = T-1$ blocks are uniform over all labelings with those counts. Hence $\widehat{\mu}_p^{HT}$ is conditionally design-unbiased for $\mu_p = B^{-1}\sum_{b=1}^{B} Y_b(p)$. Consequently,*

$$\widehat{\Delta}_F^{HT} = \widehat{\mu}_F^{HT} - \widehat{\mu}_N^{HT}, \qquad \widehat{\Delta}_I^{HT} = \widehat{\mu}_D^{HT} - \widehat{\mu}_F^{HT} \tag{6}$$

are design-unbiased for the finite-population fallback and hidden-selection effects. The total delegation effect satisfies both the estimand and estimator identities

$$\Delta_T = \mu_D - \mu_N = \Delta_F + \Delta_I, \qquad \widehat{\Delta}_T^{HT} = \widehat{\Delta}_F^{HT} + \widehat{\Delta}_I^{HT}. \tag{7}$$

For any tested pair p, q and remaining policy r, under the pairwise sharp null $Y_\ell(p) = Y_\ell(q)$ for every preterminal block ℓ, conditioning additionally on the set of blocks assigned to r leaves a uniform fixed-count randomization of p and q. The resulting pairwise Fisher test is exact regardless of the potential outcomes under r. Holm adjustment of the two primary pairwise p-values controls strong familywise error. For $\alpha \in (0,1)$, define $\rho_p = 1 - (n_p - 1)/B$ and $h_p = \sqrt{\rho_p \log(6/\alpha)/(2n_p)}$. Conditional on the same stopped design,

$$\Pr_Z\left(\bigcap_{p\in\{N,F,D\}} \{|\widehat{\mu}_p - \mu_p| \le h_p\}\right) \ge 1 - \alpha, \tag{8}$$

so every contrast $p - q$ is simultaneously covered by $[\widehat{\mu}_p - \widehat{\mu}_q - h_p - h_q, \widehat{\mu}_p - \widehat{\mu}_q + h_p + h_q] \cap [-1, 1]$. No monotonicity of fallback or delegation is required.

Under this conditional design, the Horvitz–Thompson mean and observed arm mean are identical. We report that common point estimate, use Newcombe intervals as descriptive companions, and compute exact pairwise Fisher randomization p-values. Fix a tested pair a, b, let c be the nuisance third policy, and condition on its realized assignment set. If K_{ab} of the remaining $n_a + n_b$ binary outcomes are successes, then under the pairwise sharp null the number X_a assigned to a has

$$\Pr(X_a = x \mid K_{ab}, n_a, n_b, Z_c) = \frac{\binom{n_a}{x}\binom{n_b}{K_{ab}-x}}{\binom{n_a+n_b}{K_{ab}}}. \tag{9}$$

We sum this law over the one-sided and absolute contrast tails; a 100,000-draw pairwise permutation is retained only as an implementation check. Permuting the nuisance arm would instead test the stronger global sharp null and can have the wrong size for a registered pairwise hypothesis. Independently drawing new labels with probability 1/3 would also fail to condition on the stopped design. The proof and validation protocol appear in section C.

The two directional components form one family. Their one-sided conditional randomization tests receive Holm adjustment, and their confidence columns also include Bonferroni–Newcombe intervals with 95% familywise coverage under the usual two-sample binomial interpretation. We label the latter descriptive, not randomization inversions. The release additionally reports the interval in equation (8), which is conservative but valid for the bounded finite-population randomization target without independent arms or a binomial model. The point estimate, interval, and randomization test are emitted only when every verified request has a binary outcome: `succeeded` is one, `failed` or `cancelled` is zero, and `unknown`, missing, or malformed values remain missing. Arm-level $[0, 1]$ bounds then propagate to each contrast. A second, wider sensitivity reconstructs the intended first policy from the frozen seed for every candidate block before the terminal gate block and assigns both missing/noncompliant treatment records and missing outcomes to their two worst-case endpoints. An unreconstructable policy widens the contrast to $[-1, 1]$. These are intended- assignment attrition bounds, not a per-protocol treatment effect.

6.3 Frozen gate, current support, and amendment history

The original H81 protocol releases the earliest chronological prefix with at least 40 verified first-position assignments in every arm, regardless of sign. At the outcome-free release revision `30b430e2a09` (the full immutable hash appears in the release manifest), the arm counts are 32, 24, and 28. Assignment replay and treatment-metadata checks pass for all 84 verified blocks, so outcomes remain physically unqueried by the public pre-gate analysis. When the gate opens, the analysis will exclude its terminal block and condition on the remaining arm counts; this correction was frozen before the first H81 outcome release. Commit `4d66fda`, also before release, freezes the simultaneous intervals and the outcome/treatment missingness rules above; synthetic unknown-outcome and noncompliance adversaries pass. A later outcome-blind proof audit replaced an all-arm global-sharp-null permutation with the pairwise conditional law in equation (9). Its nuisance-effect size audit and exact power surface imply that the fixed gate is informative for large wedges but not an equivalence design. A second outcome-blind amendment adds a simultaneous finite-population confidence set and exact joint-Holm power: on the frozen grid, 80% worst-terminal-policy power requires a 35-point component wedge. The candidate funnel is narrow: 74 eligibility records cover two models at ranking positions five and six; both models are eligible in every recorded run and adjacent eligible support never changes. The experiment will therefore identify a finite-sample policy effect for those model–time blocks, not a market-wide average. The remaining arm deficits are 8, 16, and 12; calendar-time forecasts are deliberately omitted because eligibility and workflow failures make the short-run accrual rate nonstationary.

H80 is a separate four-arm replication. Its original protocol also specified 40 observations per arm. After its all-position aggregate outcomes were visible, but before that original first-position gate opened, the promotion threshold was raised to 500 per arm as a conservative replication amendment. We retain the history rather than calling 500 the original preregistration. At the same pinned revision H80 has 156 verified first-position blocks with arm counts 37, 43, 38, and 38. H80 and H81 are never pooled because their policies and model support differ.

H95 addresses H81's narrow transport without modifying H81. Its candidate frontier is ranking positions 7–30, excluding the models assigned to H80 and H81; a candidate must carry a current Hugging Face identifier, expose at least two positive-price providers, and pass the public endpoint fetch. Each hourly run writes the full eligibility funnel before any purchase, samples three eligible models uniformly, and assigns each of the three H81 policies to first position exactly once. The confirmatory sample is the first 120 valid written triplet plans: 360 first-position assignments with exact 120-per-arm balance and at most 1,080 one-token requests. Missing planned requests and noncompliance are coded as structural intent-to-treat failures. A recorded compliant outcome is binary only when it is `succeeded`, `failed`, or `cancelled`; unknown or malformed values suppress complete-data inference and enter $[0, 1]$ bounds. For each elementary null, Fisher tails condition on the triplet's nuisance-policy assignment and convolve the two remaining focal- policy swaps; a 100,000-draw conditional-swap simulation is a fail-closed implementation audit. The two directional contrasts receive Holm adjustment. Paired-t intervals are descriptive and Bonferroni-adjusted as a two-contrast family; a wider Hoeffding set gives simultaneous design-valid coverage for the two bounded finite-population contrasts. Proposition 4 gives the identification result, exact pairwise law, and confidence bound.

Broad transport additionally requires at least eight audited models, effective model count five, no model over 35%, no six-hour bin over 20%, and whole-triplet leave-one-model-out sign stability. The protocol, collector, analyzer, and workflow were frozen in commit `00351dd` before the first request; the prerelease inference and telemetry audit was frozen in commit `f170d89` with the outcome gate still closed. A second proof audit corrected the pairwise reference law and interval interpretation after exact-head run `29568434722` again reported `outcomes_queried=false`. At revision `30b430e2a09`, the outcome-free support is five compliant triplets, 15 blocks, nine unique models, effective model count 7.76, perfect plan replay, and zero missing first records. The first 12 first-position rows predate row-level provider-order lengths and remain legacy-unverified; all three records in the first hardened triplet are auditable and pass, for current audit coverage 20% and conditional pass rate 100%. Three of five triplets occupy the largest six-hour bin, so this early support fails the time-transport gate. Because the three model blocks execute sequentially, a direct-policy interpretation also requires no treatment-dependent spillover from an earlier model block; the release includes a position-by-policy diagnostic. Because model order is fixed before the independent policy shuffle, the position-zero cells also support a secondary randomized estimator over the first model block, which has no earlier H95 request. It uses exact conditional pairwise tests and a simultaneous Hoeffding–Serfling interval. In a planted- carryover audit the primary estimator's absolute bias reaches 0.244 while the position-zero estimator remains within 0.001, but its mean design interval width is 0.810. Agreement at release supports but cannot prove no interference; disagreement narrows the estimand to position zero. H95 has no released outcome and is never pooled with H81.

7 Secondary evidence and falsification

The focal result is the randomized clearing decomposition. We retain the public pricing exercises only where they discipline its interpretation; full estimates, plots, theorem statements, and proofs are in the appendix.

Steering memory. Following Johnson et al. [2023], we ask whether a currently cheap provider receives additional default selection after a recent price cut. Among 251 provider–model–days, cheapest providers without a cut in the prior seven days receive 23.3% of our default selections, versus 3.9% after a cut; a model-day and current-rank controlled coefficient is also negative. Restricting to independently confirmed eligibility preserves the direction (30.0% versus 9.5%) but leaves 11 and one cells. This is a revealed-policy audit for one account, not a causal effect of cutting price. Recent cuts may proxy for poor capacity, so the data reject a positive past-cut reward on this sample but do not identify whether the rule protects quality or taxes undercutting.

Pricing technology and timing. Only 19 of 69 exposed providers reprice at least once. Conditional on model-day, slow providers quote 8.9% more than fast providers: the fast-provider log-price coefficient is -0.0854 (95% interval $[-0.1454, -0.0254]$). On the smaller public-quality sample, the coefficient is -0.2906 with interval $[-0.5493, -0.0319]$. The cadence association resembles Brown and MacKay [2023]; the mechanism does not. The frozen temporal design has only two independent waves and zero slow-initiator response comparisons. State-only versus reaction-rule holdout RMSE is 0.06856 versus 0.06204 over 82 linked reactions. The model-cluster bootstrap MSE-improvement interval is $[0.000220, 0.026124]$, but it contains only two clusters and the exact sign-flip p-value is 0.25; the registered verdict remains predictively indistinguishable. Five-minute sampling also makes latent within-bin leaders interchangeable. In the corrected frozen panel, 243 of 313 events have a unique strictly prior rival inside 72 hours, so the sharp share caused by that named rival remains $[0, 77.6\%]$, with no positive lower bound. Thus public quotes alone do not identify quote-surface front-running; the constructive result and full audit appear in section H.

Revenue-share accounting. On a common market-fixed-effect sample, the log quantity-share price coefficient is mechanically one less than the log revenue-share coefficient; residual-based standard errors are identical. In 6,414 observations, the coefficients are -1.0269 and -0.0269 with common clustered standard error 0.0933. The identity holds to 3.6×10^{-15}, and the latter coefficient is equivalent to zero inside $[-0.25, 0.25]$ ($p = 0.0084$). Yet the specification that absorbs both date–model and provider–model effects yields -0.020 (SE 0.409), unlike the published -1.02 benchmark; only 50 listed-price moves and 0.157% of raw centered price variation survive both effects. The result corrects the accounting interpretation of a minus-one share coefficient. It does not identify a causal demand derivative, revenue optimum, or welfare gap. The theorem, replication ladder, and support audit are in section G.

Cross-router catalogs. The first simultaneous Glama–Requesty–NemoRouter cross-section contains 784 quotes and 29 Hugging-Face-linked exact provider–model pairs. Twenty-eight pairs have identical input and output prices (96.6%, Wilson interval $[82.8\%, 99.4\%]$); the one DeepSeek pair has a 296.6% registered-workload wedge. This is consistent with routers redistributing an upstream price book, but one cross-section cannot identify pass-through or staleness. H94 is now prospectively active with a future-only cutoff of 2026-07-17 04:30:20 UTC; the discovery cross-section is excluded. One prospective snapshot per router is eligible, but there are zero prospective transitions or matched shocks, so no dynamic result exists. Moreover, 28 of the 29 simultaneous pairs have at least one observed contract-term conflict, and region, SLA, rate limit, capacity, fallback, billing, cache, and priority terms remain partly unobserved. Exact token prices therefore do not establish contract equivalence. The dynamic gates require seven days, 48 snapshots per router, 30 transitions, 15 common shocks, and ten changing commodities. The full plot and protocol summary are in section F.

Other diagnostics. Entry, retry, and event-shape screens do not clear their identification bars. The provider-count elasticity with respect to tokens is 0.149 (95% interval $[0.126, 0.175]$), numerically close to a long-memory benchmark but confounded by simultaneous entry and demand. Demand-controlled retry OLS is 0.167, whereas the cross-model rate-limit IV estimate is 0.023 with interval $[-0.043, 0.085]$. Finally, reconstructing the initiating provider's next action splits apparent punish-and-revert events almost evenly between candidate punishment (42.1%) and cuts the initiator later withdraws (41.1%). These are useful falsifications of simple stories, not promoted evidence of entry efficiency, retry externalities, or collusion.

8 Design implications, not welfare estimation

The evidence identifies a missing contracting margin, not a structural welfare number. A user or harness values a policy through its attempted-request spend, completion probability, delay, and fidelity. H81 can compare the first three for one owned account once its gate opens; it does not observe user value, provider cost, external congestion, or task-level quality. Transfers also affect provider entry and investment even when they cancel from a planner's static accounting. We therefore do not estimate a welfare loss or claim that a particular router policy is efficient.

The narrow implication is still useful. A price-only best-execution rule is incomplete when the displayed quote is not an execution commitment. For a fixed workload, an auditable procurement score must combine expected spend with failure, delay, and fidelity terms, and must state whether fallback is included. Raw token price implements that score only when the nonprice terms are equal across eligible providers or already embedded in enforceable contingent prices. H81 and its independent fixed-horizon H95 replication test whether fallback and delegated selection are empirically relevant pieces of this score; neither estimates their social weights.

Execution-contingent prices, explicit fallback prices, and provider-level fill rates would make the displayed and clearing layers more congruent. Capacity certificates are one possible implementation, studied in a companion mechanism paper. The stylized free-entry benchmark from the earlier manuscript is retained with its proof in section E; none of its primitives is calibrated here.

9 Related work

Model routing. FrugalGPT, RouteLLM, and RouterBench study how to choose among models with different quality and cost Chen et al. [2023], Ong et al. [2024], Hu et al. [2024]. Their unit is a model. Our unit is a provider-model endpoint, holding the model label fixed and measuring the market that supplies it. The two problems compose: a harness may choose the model and a marketplace may then choose the supplier.

Pricing technology and focal points. Calvo [1983] provides the constant-hazard benchmark; observation and menu-cost models motivate duration, gap, and change-size diagnostics [Alvarez et al., 2011, 2016]. Brown and MacKay [2023] show that heterogeneous repricing frequency can soften competition. Knittel and Stango [2003] identify a nonbinding regulatory ceiling as a focal point. We import their empirical logic and then show why an adjacent-level counterfactual is insufficient in a market with shared discrete menus. Exact endpoint-label and matched common-menu nulls separate the existence of a price atom from claims about anchor identity or strategic following. The latter is a dynamic pricing analogue of the reflection problem: correlated behavior and endogenous response are inseparable without reference-group or shock restrictions [Manski, 1993].

Platform steering and marketplace experiments. Johnson et al. [2023] connect prominence rules to algorithmic pricing. Our steering-memory statistic is an empirical implementation of their sign restriction. Switchback and marketplace experiments require special care because treatment changes congestion and later outcomes Bojinov et al. [2023], Holtz et al. [2025]; our first-position estimator protects the firmness contrast from arbitrary within-block carryover. Online matching under stochastic rewards provides the OR analogue for provider selection Brubach et al. [2025], but the inference setting adds sticky provider menus and hidden eligibility.

Two-sided procurement and entry. The router brings harness demand and provider capacity onto the platform, but it also evaluates price jointly with latent service attributes. This combines the price-allocation problem of two-sided platforms [Rochet and Tirole, 2003] with the multi-attribute scoring logic of procurement [Che, 1993]. Our entry proposition specializes Mankiw and Whinston [1986]: an entrant both adds a redundant delivery path and captures jobs that incumbents would have served. Unlike a standard homogeneous-product application, the positive social margin is the probability that the entrant is available exactly when every incumbent is not.

Emerging AI-platform evidence. Recent work studies quality-adjusted AI prices and platform self-preferencing Wang [2026], Zhang et al. [2026]. Demirer et al. [2025] use OpenRouter and Azure data to document supply, demand, and pricing and estimate an open-source provider–model prompt-price elasticity near minus one. That estimate is the closest benchmark for provider revenue stationarity. We show that any common market-fixed-effect quantity-share coefficient is mechanically one less than the corresponding revenue-share coefficient, then reproduce the published fixed-effect ladder and audit the sparse repricing support. The accounting result is exact; the current within-provider/model discrepancy is promotion-gated because a first-order condition requires a causal own-price response. We complement these studies with provider-level microstructure and a reproducible realized-policy audit. The institutional APIs are documented by the platforms themselves OpenRouter [2026], Hugging Face [2026]; all policy claims in this paper are versioned to the captured interfaces rather than generalized to undocumented behavior.

10 Limitations and conclusion

Five limitations define the claim boundary. First, the main pricing panel is nine days, so dynamic hazard and cadence results await the registered 30-day both-vintage re-estimation; the identity and lagged-landing hard nulls are post-freeze and await the same replication, while the local outcome-free calendar contains 10/30 dates. The matched-menu test controls one-sided size in the conditional simulation but has only 43.4% power at the observed effect and can be far more conservative in known-clock panels; its nonrejection is not evidence of zero response. Exact cluster sign-flip p-values are 0.425 and 0.214, and every registered empirical calibration diagnostic rejects the known-clock family as a fitted description. A separately preregistered 15/15-date whole-market holdout test remains unopened until the 30-date gate. Second, H81 remains below its original release gate and spans only two repeated models. At its minimum preterminal counts, exact Bernoulli scenarios require a 25–35 percentage-point effect for 80% power at the conservative familywise threshold; a null result would therefore leave modest economically relevant effects unresolved. Exact joint-Holm enumeration puts worst-terminal-policy 80% power at a 35-point component effect, while the model-free simultaneous design interval has mean width about 0.76 in fixed-schedule stress tests. H80's later replication-promotion gate remains at 500 per arm, versus current outcome-free first-position counts of 37, 43, 38, and 38. H95 broadens prospective support, but has no released outcome and cannot retroactively repair H81. Even at its 120-triplet horizon, the simultaneous bounded-outcome radius is 0.270 per primary contrast; tighter paired-t intervals require a triplet-superpopulation interpretation. Its first 12 rows predate the expanded provider-control metadata, and direct-policy language requires no treatment-dependent carryover across its sequential model blocks; the identified position-zero sensitivity removes earlier within-triplet carryover but has mean design width 0.810 and cannot detect interference from unrelated prior traffic. Both limitations remain visible in the release diagnostics. Third, owned probes identify one account, region, prompt family, and time window; they do not identify market-wide provider shares. Fourth, public operational fields are router-published aggregates with missingness, not raw attempt logs. Fifth, provider cost, private rebates, task fidelity, and cross-user request order are unobserved. We therefore do not claim literal front-running, collusion, private cost, or a welfare loss. The generalized-cost score and free-entry proposition are conditional mechanism benchmarks, not calibrated estimates: their value, cost, setup-cost, and deliverability primitives are precisely among the missing objects.

Within those limits, the evidence changes the economic description of inference markets. Prices are not the rapidly clearing state variable. Providers post sticky administered menus, minimum-price ties exceed independent grid benchmarks, but exact atoms do not survive hard tests of author identity. Strategic following is neither promoted nor ruled out: its hard null is observationally defensible but underpowered at the observed effect. The outcome-free H81 design can distinguish fallback from hidden selection once its fixed gate opens, and H95 supplies a separate fixed-horizon multi-model replication; neither has a promoted outcome now. The platform's allocation-memory audit does not reward recent undercutting in the observed panel. Even within these boundaries, the inference router is a market designer: its admission, fallback, and steering rules determine the delivered product as much as the visible price surface does.

The resulting research agenda is concrete. Longer panels test whether gap-driven repricing and matched-menu excess survive new vintages. The randomized H81 gate and independent H95 horizon estimate the deliverability frontier without order confounding. Provider- or router-side eligibility logs would separate rational quality protection from a tax on undercutting. Transparent decentralized markets supply an external laboratory in which accepted bids and termination reasons are public. Together, these designs can turn a market that currently displays menus into one whose commitments, allocation, and welfare can be audited.

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A Price-atom diagnostics, nulls, and proofs

A.1 The tie atom and its null

For each model-day with at least two providers, order the completion quotes and define the relative second gap

$$g_{md} = \frac{q_{(2),md} - q_{(1),md}}{q_{(1),md}}. \tag{10}$$

Across 1,311 model-days, $g_{md} = 0$ in 45.9%. A zero gap is especially common in two-provider markets, but the atom persists in thicker markets. Among all non-minimum endpoint observations, 47.4% are exactly tied at the minimum; the remaining distribution has relatively little mass in very small undercuts.

Rounding can generate ties mechanically. We therefore construct an independent grid null. Within each model-day, each provider draws a log-price deviation from the pooled empirical distribution of deviations from model-day medians. Draws are independent across providers, added to the observed model-day median, and snapped to the observed cent-per-million-token grid. We preserve the observed number of providers by model-day and repeat the procedure. The null produces a 13.4% mean tie rate with 1.0 percentage-point simulation standard deviation. Snapping to a coarser dime grid increases the null to 27.6%, still well below 45.9%. Because the pooled deviation distribution retains some empirical tie mass, this null is conservative with respect to independent coordination on the same grid point.

Post-freeze robustness (1,456 model-days, 148 model clusters) retains a 45.1% tie rate. The cent-grid null is 12.9%, giving a 32.2-point excess (model-cluster bootstrap 95% interval [25.8, 37.8]); the dime-grid null is 27.2%, giving an 18.0-point excess [9.7, 25.1]. The bootstrap resamples whole models and re-simulates the null within each draw; it is a robustness refresh, not a replacement for the frozen v3 vintage.

Statistic	Estimate	Denominator
Observed tie at minimum	45.9%	1,311 model-days
Independent cent-grid null	13.4%	matched simulations
Independent dime-grid sensitivity	27.6%	matched simulations
Observed / cent-null ratio	3.4	–
Observed / dime-null ratio	1.7	–

Table 5: Minimum-price tie atom. The null preserves market thickness, observed deviation heterogeneity, and a discrete price grid, but breaks cross-provider dependence.

A.2 Identifying the focal level

The selected-tie identity statistic needs its own null. If a model has n providers, t are tied at the minimum, and r author-operated labels are placed exchangeably among provider slots, then the conditional probability that an author label lies in the tied set is

$$1 - \frac{\binom{n-t}{r}}{\binom{n}{r}}. \tag{11}$$

It equals one when all providers tie. Thus conditioning on a tied model can make an author match nearly automatic. In the post-freeze audit of the frozen panel's final snapshot, the robust author-family crosswalk produces 45 matches among 50 selected tied models, or 90.0%. Summing equation (11) across those markets predicts 45.97 matches, or 91.9%; the exact Poisson-binomial upper-tail probability is 0.913. The previously emphasized 65-of-72 frozen statistic is therefore demoted: its high level does not identify the author's price as focal.

The all-market estimand avoids selecting on a tie. It starts from every multi-provider model with one author-operated and at least one third-party endpoint. At the primary dime-per-million-token spacing, 51 of 94 models, or 54.3%, have a third-party quote exactly at the author price, versus 1.5% at 40 adjacent placebo levels. The resulting 52.8-point contrast is precisely estimated but answers only whether an observed price level contains mass. Empty nearby grid points are not an identity null when providers use a common discrete menu.

The harder null preserves each model's entire realized price multiset and places the author label uniformly among its endpoint slots. An endpoint is "shared" when at least one other endpoint has the same price. If s_m of n_m endpoints are shared, the random-anchor probability is s_m/n_m. Authors occupy a shared price in 54.3% of models, while the summed random-anchor expectation is 50.26 of 94, or 53.5%. The exact Poisson-binomial upper-tail probability is 0.466. The mean author-minus-random-anchor difference is 0.79 points with a 20,000-draw author-cluster interval of [-9.0,16.1] points and leave-one-author-out range [-1.9,6.5]. Comparing author–third-party equality with equality among third parties gives -0.94 points over 48 eligible models, with interval [-8.8,12.0]. Thus the atom is real, but author identity is not special under a price-multiplicity-preserving null.

Identity statistic	Estimate	Benchmark
Selected-tie author match	90.0%	91.9% random label
All-market author at shared price	54.3%	53.5% random endpoint
Author-minus-random endpoint	0.79 pp	[-9.0,16.1] cluster interval
Author-vs-third-party pair density	-0.94 pp	[-8.8,12.0] cluster interval

Table 6: Price-multiplicity-preserving identity audit. Both conditional and all-market author statistics fail their exact endpoint-label nulls. The post-freeze specification is fixed for the 30-date release.

The same problem arises in time series. We isolate consecutive snapshots no more than 15 minutes apart in which exactly one provider changes price and all compared quotes are observed on both sides. Of 196 revisions across 18 models and 15 providers, 40, or 20.4%, land exactly on a rival's strictly prior quote. Only 1.0% land on the average of 40 adjacent placebo levels, producing a seemingly large 19.5-point excess. But common price menus can make the adjacent levels empty by construction.

For each event, our matched-menu null draws the same number of prices as observed rivals from other models in the immediately prior snapshot, restricted to prices within a factor 1.25 of the mover's new quote. The exact hypergeometric hit probability preserves local price scale, risk-set size, and the global frequency of each exact menu point. It predicts 13.4% matches. The remaining 7.0-point excess has model-cluster interval [-23.1,12.2] and provider-cluster interval [-22.4,18.2]; its leave-one-model-out range is [-10.4,9.7]. One model supplies 73.5% of events and 85.0% of exact landings. Wider control bands mechanically lower the menu baseline, so the registered replication retains the most local 1.25-band null. The current panel does not distinguish strategic following from a common asynchronously refreshed menu. A past-only same-model sensitivity with a 24-hour washout is more favorable: 189 events have a 6.8% historical-menu baseline and 14.4-point excess. Yet its model-cluster interval is [-0.1,17.5] points, while its positive provider-cluster interval is driven by the same concentrated models and movers. We register this SKU-specific comparison as secondary rather than selecting it over the tougher primary null.

The preregistered same-provider/across-model control was committed before its estimate was computed. Among 175 comparable events, only 12, or 6.9%, reuse the mover's own price on another model at the strictly prior timestamp. Own-menu support is 5.3% among exact rival landings and 7.3% otherwise, a −2.0-point association with model-cluster interval [-31.0,0.9]. Removing those 12 events leaves 163 own-menu-novel revisions: their exact landing rate is 22.1% against an 8.6% matched-menu benchmark. The 13.4-point residual is larger, and every leave-one-model-out estimate is positive [1.4,15.2], but its model-cluster interval [-11.5,18.5] still fails the frozen promotion rule. The negative control therefore weakens a simple provider-template explanation without identifying rival response.

Lagged-landing statistic	Estimate	Benchmark or interval
Isolated continuous revisions	196	18 models, 15 providers
Exact landing on lagged rival quote	20.4%	40 events
Adjacent-grid placebo	1.0%	19.5 pp excess
Matched common-menu probability	13.4%	factor-1.25 risk set
Exact-minus-common-menu	7.0 pp	[-23.1,12.2] model-cluster interval
Past-only same-model menu	14.4 pp	[-0.1,17.5] model-cluster interval
Own-provider cross-model support	6.9%	−2.0 pp landing association
Own-menu-novel common-menu excess	13.4 pp	[-11.5,18.5] model-cluster interval

Table 7: A weak placebo suggests price following; the matched common-menu null does not. The table is an exploratory frozen-panel audit registered unchanged for the earliest 30-date release.

Proposition 2 (Asynchronous-menu observational equivalence)**.** *Let a finite snapshot panel take values in a discrete public price menu. Suppose one-provider revisions can be scheduled by provider-specific refresh clocks and selection rules driven by a latent public menu state, without dependence on rival quotes. A model that instead permits rival-triggered revision is also admissible. Then any observed exact cross-sectional match or exact landing on a strictly prior rival quote can be generated by both models. If L of N isolated revisions land exactly, the share of all revisions that are exact strategic landings is sharply identified only within* $[0, L/N]$.

The constructive proof is in section H. For the frozen panel, $L/N = 40/196 = 20.4\%$. The matched-menu test restricts the otherwise unrestricted null by imposing exchangeability within a local public risk set; its failure to reject is empirical, while proposition 2 explains why the weak adjacent-grid result cannot carry a behavioral interpretation.

The hard null can also fail by being too conservative. Let p_e denote an event's nonreactive exact-landing probability and q_e its matched-menu benchmark. Suppose a response, when it occurs, forces an exact rival-price landing.

Proposition 3 (Detection threshold of a dominating menu null)**.** *If reactive replacement occurs with probability ρ, independently conditional on the event risk set, then*

$$\mathbb{E}[Y_e - q_e] = p_e - q_e + \rho(1 - p_e). \tag{12}$$

When $q_e \geq p_e$, the matched-menu residual has correct one-sided sign under the nonreactive mechanism but becomes positive only if

$$\rho > \rho_e^\star \equiv \frac{q_e - p_e}{1 - p_e}. \tag{13}$$

For a common ρ over a fixed event panel, replace (p_e, q_e) by their event means. Thus conservativeness and power loss are two implications of the same benchmark dominance.

The frozen-design simulation holds each event's q_e and model cluster fixed. In 1,000 replications per response level, one-sided false promotion is 3.5%; power is 29.7% at $\rho = 0.05$, 51.9% at 0.10, and 94.7% at 0.25. The observed 7.0-point excess corresponds to $\rho = 0.081$ in that conditional mixture and only 43.4% interpolated power. A second experiment generates 250 complete 576-tick panels per level from public menus and known provider refresh clocks. At $\rho = 0$ the exact-landing rate is 59.9% while the global-menu benchmark is 90.4%; the implied fixed-panel threshold is approximately 0.762. The unchanged test never promotes, even at $\rho = 0.50$. This is not evidence that the market has a large or small ρ; it shows that failure to reject the hard null cannot be read as evidence of no strategic behavior.

Two preregistered robustness disclosures sharpen that boundary. Exact enumeration of all 2^{18} model-cluster sign assignments gives one-sided $p = 0.425$ for the full residual; enumeration over the 14 own-menu-novel clusters gives $p = 0.214$. The positive event-weighted estimates therefore lack broad cluster support. The known-clock family is also not empirically calibrated. Its null 5th–95th percentile ranges are 1,228–3,381 events, 49.0–73.2% exact landings, 89.1–91.3% menu probabilities, and −41.5 to −17.3 points of residual. The empirical values—196, 20.4%, 13.4%, and +7.0 points—fall outside all four ranges. Accordingly SIM2 is a stress-test construction showing possible benchmark dominance, not a fitted data-generating process or an estimate of ρ.

The failed calibration is informative about the next test rather than a reason to retune SIM2. Before the 30-date panel exists, we preregistered a 15-date training and 15-date holdout comparison that matches whole contemporaneous markets on endpoint count, price scale, dispersion, duplicate-price mass, and minimum ties. It fits a single excess-landing parameter on training events and evaluates both the matched-market residual and out-of-sample log-score gain on the untouched holdout. The analyzer cannot load prices or construct events before the calendar gate. Its support and promotion rules appear in section I. This design can reject a declared market-exchangeability benchmark; it still cannot identify intent or private request order.

Calibration statistic	Empirical	SIM2 p05	SIM2 median	SIM2 p95
Extracted events	196	1,228	2,150	3,381
Exact landing share	20.4%	49.0%	58.8%	73.2%
Matched-menu probability	13.4%	89.1%	90.5%	91.3%
Exact-minus-menu residual	+7.0 pp	−41.5 pp	−31.5 pp	−17.3 pp

Table 8: Frozen empirical statistics against the registered $\rho = 0$ SIM2 distribution. Every empirical value is outside the simulated 5th–95th percentile range; no simulation parameter was retuned.

Direct provider APIs and router quotes are at exact parity in 99.6% of matched cases; we see almost no systematic direct-channel undercut. Of 31 isolated tie formations, 74.2% are formed by a

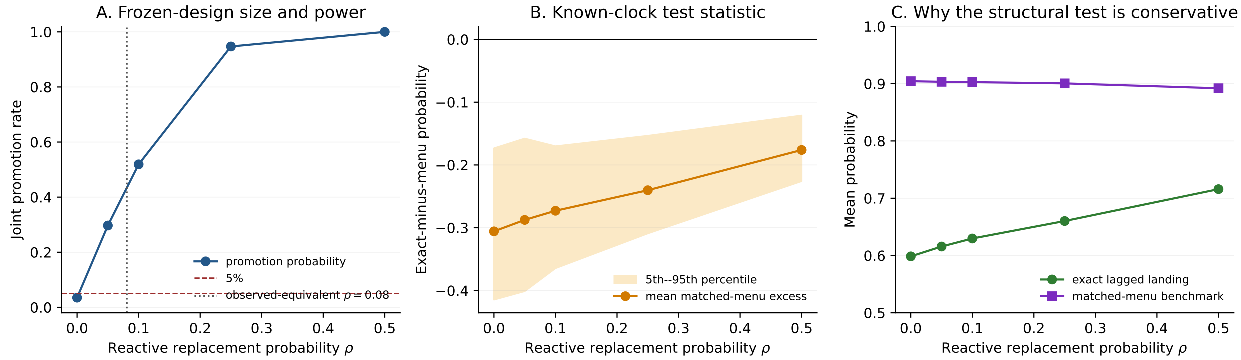

Figure 1: Preregistered size and power. Panel A conditions on the frozen 196-event design. Panels B–C run the unchanged statistic on known-clock asynchronous-menu panels with nested reactive replacement. The global-menu benchmark controls false positives but can dominate the focal rival set enough to eliminate power. The registered mechanism fails the empirical calibration audit and is interpreted as a stress test.

downward move, and 88.4% of 43 isolated tie breaks are downward. Those directions are competitive. The evidence supports shared discrete price menus, not a named anchor: common cost conventions, provider templates, strategic matching, and a Schelling-like device remain observationally compatible.

B Variable construction and data-generating process

Variable	Construction
Completion quote	Output-token price in dollars per million tokens for the standard endpoint variant
Price event	First five-minute capture whose quote differs from the preceding same-endpoint capture
Cheapest rank	Dense rank of the workload-specific quote within model and capture time
Tie at minimum	At least two distinct providers have the exact minimum quote after common-unit conversion
Quote age	Time since the last observed event; first panel spell is left-censored
Recent cut	At least one negative completion-price event in the preceding seven calendar days
Default selection share	Selected-provider count divided by default probes in the provider-model-day cell
Rate-limit share	Router-published rate-limited count divided by successful plus rate-limited counts in the published window
Author-operated endpoint	Provider name matches the model author under the checked author-provider crosswalk
Direct parity	Direct API quote equals the contemporaneous router quote within machine tolerance

Table 9: Key variable definitions.

The capture scheduler targets five-minute cadence, but network and source failures create irregular intervals. All duration variables use actual UTC timestamps. Missing captures are never forward-filled for event timing. A quote that is unchanged across a gap contributes exposure but cannot locate an unobserved intermediate round trip; historical sources are therefore used as a lifecycle sensitivity, not to impute high-frequency paths.

The platform's five-minute and thirty-minute counters can overlap mechanically. Regressions use one horizon at a time. Capacity ceilings are displayed metadata, not contractual commitments, and missing ceilings are not coded as infinite. Derank transitions and rate-limit onsets are router enforcement events; they do not reveal which user's request caused the event.

C Proof and validation of the delegation decomposition

Proof of proposition 1. Condition on the stopping time T, terminal policy, preterminal counts, and the finite potential-outcome schedule $\{Y_b(N), Y_b(F), Y_b(D)\}_{b=1}^{B}$, where $B = T-1$. Because the terminal policy has count 39 at $T-1$, the stopping event imposes no further ordering restriction after we condition on the preterminal counts. The labels on the first B blocks are therefore a uniform fixed-count permutation. In particular, for each b and p, $\mathbb{P}(Z_b = p \mid T, n_N, n_F, n_D) = n_p/B = \pi_p$. Consistency gives $Y_b^{obs} = Y_b(p)$ whenever $Z_b = p$, so

$$\mathbb{E}_Z\left[\frac{\mathbf{1}\{Z_b = p\}Y_b^{obs}}{\pi_p}\middle| T, n_N, n_F, n_D\right] = \frac{\pi_p Y_b(p)}{\pi_p} = Y_b(p).$$

Summing over blocks and dividing by B yields $\mathbb{E}_Z[\widehat{\mu}_p^{HT}] = \mu_p$. Linearity of expectation proves design-unbiasedness of both component contrasts. Finally,

$$(\mu_F - \mu_N) + (\mu_D - \mu_F) = \mu_D - \mu_N$$

and the same cancellation holds pathwise after replacing each μ_p by $\widehat{\mu}_p^{HT}$. The argument uses neither an ordering of the three potential outcomes nor independence of outcomes across blocks. It requires only positive conditional assignment probabilities, correct assignment replay, and no interference into the first request from a prior request in the same block. Excluding the terminal block is essential: conditional on stopping, its assignment is fixed to the last arm to reach 40 and has zero probability under the other policies.

For the testing statement, fix policies p, q and let r denote the third. Condition further on the realized nuisance set $\mathcal{C}_r = \{\ell : Z_\ell = r\}$. Every labeling of the complement with n_p labels p and n_q labels q has the same conditional probability, because the original preterminal labeling was uniform at its three fixed counts. Under the pairwise sharp null $Y_\ell(p) = Y_\ell(q)$, the observed binary outcome on every block outside $\mathcal{C}_r$ is invariant to those relabelings. If K_{pq} of those outcomes are one, exactly $\binom{n_p}{x}\binom{n_q}{K_{pq}-x}$ labelings assign x successes to p, out of $\binom{n_p+n_q}{K_{pq}}$ labelings in total. This proves equation (9); summing its upper or absolute tail gives a super-uniform exact p-value. The argument places no restriction on $Y_\ell(r)$ because both $\mathcal{C}_r$ and its labels are held fixed. Each primary p-value is therefore valid under its own null even when the other null is false, and Holm's step-down argument gives strong familywise control without a dependence assumption.

For the interval statement, conditional fixed-count uniformity also makes the n_p blocks labeled p a simple random sample without replacement from the B fixed numbers $\{Y_b(p)\}_{b=1}^{B} \subset [0,1]$. Hoeffding's sampling-without-replacement inequality therefore gives $\Pr_Z(|\widehat{\mu}_p - \mu_p| > h_p) \leq 2\exp(-2n_p h_p^2/\rho_p) = \alpha/3$; see Hoeffding [1963], Serfling [1974]. A union bound over N, F, D proves equation (8). On that joint event, the triangle inequality gives $|[(\widehat{\mu}_p - \widehat{\mu}_q) - (\mu_p - \mu_q)]| \leq h_p + h_q$ for every pair simultaneously. Intersecting with the logical outcome-difference range $[-1, 1]$ cannot reduce coverage. $\square$

The theorem has five distinct validation targets. First, assignment replay must recover the recorded first policy from every 64-bit seed and the treatment metadata must implement the stated candidate-set and fallback controls. Second, a finite-population Monte Carlo must hold a potential-outcome table fixed, run the production assignment algorithm through the balance stopping time, and verify both the bias of the uncorrected terminal-inclusive estimator and the unbiasedness and nominal size of the conditional preterminal estimator. Third, a nuisance-arm stress test must impose each pairwise sharp null while allowing the untested policy an arbitrary effect. Fourth, fixed-schedule interval coverage and exact joint-Holm planning power must expose both the guarantee and precision cost of the 40-per-arm gate. Fifth, adversarial missingness simulations must confirm that unknown outcome codes are never coerced to failure; incomplete binary outcomes must suppress point inference and enter $[0, 1]$ bounds. The same audit reconstructs intended first-policy assignments and sends missing or noncompliant treatment records to both worst-case endpoints. Spend, latency, and provider identity are either fully observed or reported with their protocol-valid worst-case bounds. Passing these checks validates the estimator implementation; it does not solve transport from the two-model collection support to the broader marketplace.

We ran 20,000 stopped assignments at the production 40-per-arm gate. Across the fallback, selection, and total contrasts, corrected bias is respectively -1.04×10^{-4}, -0.05×10^{-4}, and -1.09×10^{-4}; each is smaller than its Monte Carlo uncertainty. The terminal-inclusive fallback estimator has bias -3.95×10^{-4} (Monte Carlo standard error 1.23×10^{-4}). The numerical magnitude is small because only one block is contaminated, but its sign and size depend on the unobserved time path and therefore cannot be assumed away. In 2,000 heterogeneous sharp-null experiments, the fixed-count test rejects at 5.05% (Monte Carlo standard error 0.49 percentage points). The pre-release implementation test then replaces a verified success with an unknown code and confirms that the point contrast and randomization p-value disappear while the true contrast remains inside the reported bounds. A second adversary corrupts one treatment-control record, accrues a valid replacement, and confirms that the omitted record reappears in the intended-assignment worst-case interval. These tests were committed in `4d66fda` before any H81 outcome release. Commit `55b5087`, also before release, replaced the published Monte Carlo tail approximation with a finite exact law. A second proof audit then found that permuting all three labels imposed an unregistered global sharp null. The corrected implementation conditions on the nuisance arm and uses equation (9); all six unique assignments in a four-block 2/2 pair fixture agree with brute-force enumeration to machine precision. In 2,000 stopped nuisance-effect experiments per primary null, the pairwise test rejects at 3.45% for fallback and 3.60% for hidden selection, whereas the superseded all-arm law rejects at 5.65% and 6.70%. Exact Bernoulli scenario power at the minimum preterminal counts 39/40 reaches 80% only for effects of 22.5–30 percentage points unadjusted and 25–35 points at the Bonferroni 2.5% threshold, depending on baseline success. Exact joint-Holm enumeration over all three possible 39/40/40 terminal-arm identities requires a 35-point fallback-only, selection-only, or equal-component wedge for at least 80% worst-case power on the five-point grid; mixed-null false rejection is at most 3.23%. Across five fixed binary schedules and 3,000 stopped assignments per schedule, worst marginal Newcombe coverage is 94.67% and worst Bonferroni–Newcombe family coverage is 95.13%. The worst design-Hoeffding–Serfling family coverage is 99.93%, but its mean contrast width is about 0.76; its formal guarantee comes from equation (8), not that simulation. Thus H81 can detect large wedges but cannot interpret nonsignificance as equivalence. The 100,000-draw pairwise tail remains solely a discrepancy audit, and the release fails closed above one percentage point. The full repository suite has 563 passes.

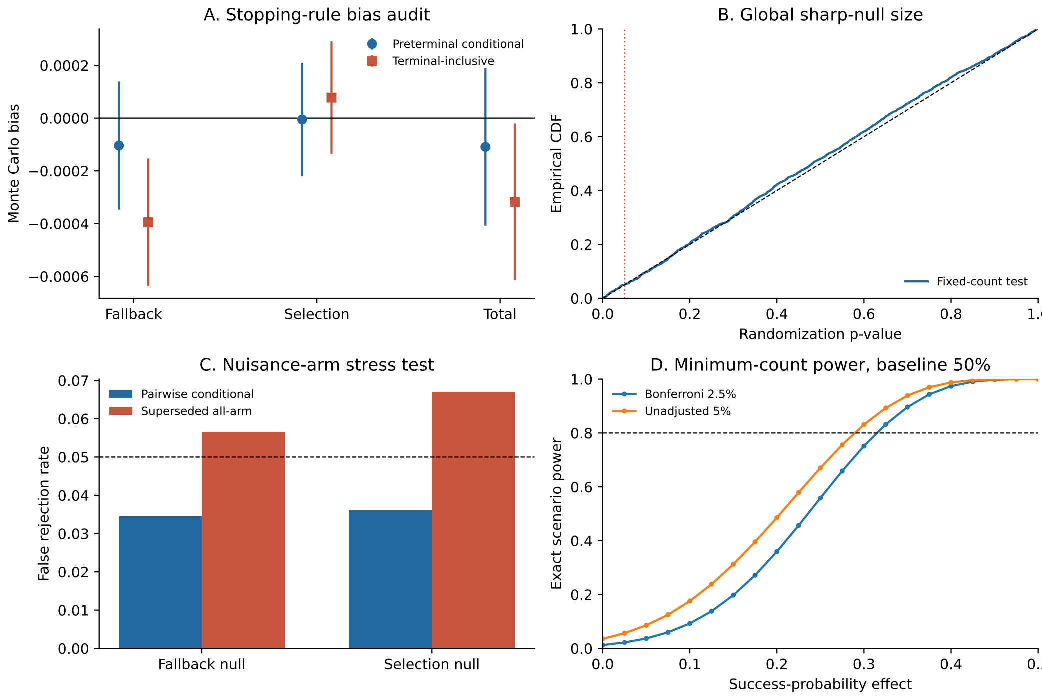

Figure 2: Validation of the stopped H81 design. Panel A compares Monte Carlo bias under a fixed, policy-specific time-varying potential-outcome schedule; bars are 95% Monte Carlo intervals. The corrected preterminal estimator is centered at zero, whereas including the gate-hitting block creates a small but design-dependent bias. Panel B shows the randomization-p empirical CDF under a heterogeneous global sharp null; the 5% rejection rate is 5.05%. Panel C holds each focal pair at its exact sharp null while giving the nuisance policy a large effect: only the corrected pairwise law controls size. Panel D reports exact model-based power at counts 39/40 and a 50% baseline. These are design validation and planning calculations, not empirical routing outcomes.

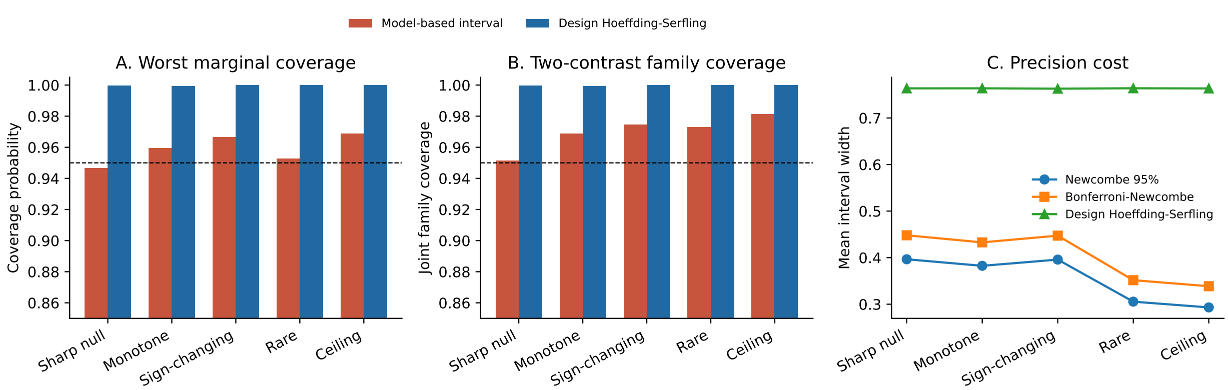

Figure 3: Outcome-blind interval audit over five fixed binary potential-outcome schedules and 3,000 stopped assignments per schedule. Panel A takes the worse coverage of the two primary contrasts; Panel B reports simultaneous family coverage; Panel C shows the precision cost. Bonferroni–Newcombe is descriptive under a binomial model. The wider design-Hoeffding–Serfling interval is valid for the conditional finite population; its observed near-100% coverage is an audit, not the source of its guarantee.

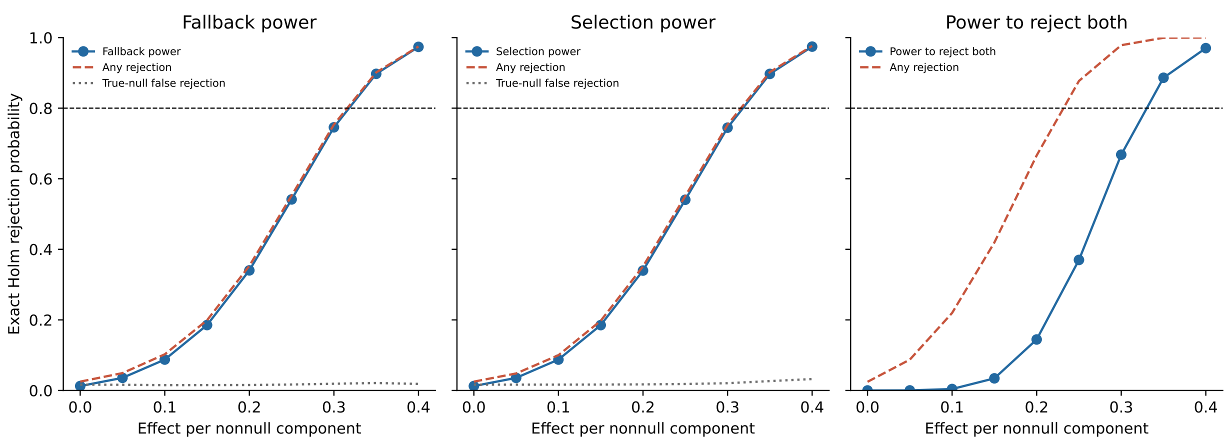

Figure 4: Exact Bernoulli-scenario power for the registered two-test Holm family at minimum preterminal counts 39/40/40, minimized over the identity of the terminal arm. The first two panels activate one component and hold the other null; the dotted line is false rejection of that remaining true null. The last panel activates equal fallback and selection components. This is a planning calculation, not an H81 outcome.

D Fixed-horizon H95 identification and exact inference

Let $J = 120$ index the prospectively written triplets and let $m \in \{1, 2, 3\}$ index the three sampled model blocks in triplet j. Write $\mathcal{P} = \{N, F, D\}$ for the no-fallback, fallback, and delegated policies. The assignment Z_j is a uniformly random bijection from the three model blocks to $\mathcal{P}$, independently across triplets. For a binary protocol outcome $Y_{jm}(p)$, define the realized-support policy mean and its estimator by

$$\mu_p = \frac{1}{3J} \sum_{j=1}^{J} \sum_{m=1}^{3} Y_{jm}(p), \qquad \widehat{\mu}_p = \frac{1}{J} \sum_{j=1}^{J} \sum_{m=1}^{3} \mathbf{1}\{Z_{jm} = p\} Y_{jm}^{obs}. \tag{14}$$

Exactly one term contributes per triplet to $\widehat{\mu}_p$.

Proposition 4 (Fixed-horizon identification, pairwise Fisher law, and confidence set)**.** *Condition on the first $J = 120$ valid plans and their sampled models. Suppose (i) each logged policy permutation is independent of the finite potential-outcome schedule, (ii) consistency holds, and (iii) a model block's outcome is not changed by the policies assigned to other model blocks in its triplet. Execution position and arbitrary position-only drift may be part of the fixed unit schedule. Then $\widehat{\mu}_p$ in equation* (14) *is design-unbiased for μ_p. Consequently, $\widehat{\mu}_F - \widehat{\mu}_N$ and $\widehat{\mu}_D - \widehat{\mu}_F$ are unbiased for the realized-support fallback and hidden-selection effects, and their sum equals $\widehat{\mu}_D - \widehat{\mu}_N$ pathwise.*

For an elementary contrast $p - q$, let r be the nuisance third policy and condition within every triplet on the model block assigned r. Under the pairwise Fisher sharp null $Y_{jm}(p) = Y_{jm}(q)$ for every j, m, let $d_{j,pq}$ be the observed outcome under p minus that under q. The exact conditional local law is

$$G_{j,pq} = \frac{1}{2}\delta_{d_{j,pq}} + \frac{1}{2}\delta_{-d_{j,pq}}. \tag{15}$$

The exact randomization law of the unnormalized contrast $J(\widehat{\mu}_p - \widehat{\mu}_q)$ is the convolution

$$G_{pq} = G_{1,pq} * G_{2,pq} * \cdots * G_{J,pq}. \tag{16}$$

For binary outcomes each local law is supported on $\{-1, 0, 1\}$, so summing the appropriate upper or absolute tail of equation (16) *gives an exact pairwise Fisher p-value even when policy r has arbitrary effects.*

Finally, define the observed triplet contrast $D_{j,pq} \in [-1, 1]$ and $\tau_{pq} = \mu_p - \mu_q$. For K prespecified contrasts, set

$$h_{J,K}(\alpha) = \sqrt{\frac{2\log(2K/\alpha)}{J}}. \tag{17}$$

Then the K intervals $[\widehat{\mu}_p - \widehat{\mu}_q \pm h_{J,K}(\alpha)] \cap [-1, 1]$ cover their finite-population contrasts simultaneously with probability at least $1 - \alpha$. For the two primary H95 contrasts at $J = 120$ and $\alpha = 0.05$, $h_{120,2} = 0.27025$.

Proof. For every fixed j, m, p, uniform bijective assignment gives $\mathbb{P}(Z_{jm} = p) = 1/3$. Under consistency and no treatment-dependent cross-model interference,

$$\mathbb{E}_Z\Big[\mathbf{1}\{Z_{jm} = p\} Y_{jm}^{obs}\Big] = \frac{1}{3} Y_{jm}(p).$$

Summing over the three model blocks and J triplets, then dividing by J, yields $\mathbb{E}_Z[\widehat{\mu}_p] = \mu_p$. Linearity proves the two component contrasts, and $(\widehat{\mu}_F - \widehat{\mu}_N) + (\widehat{\mu}_D - \widehat{\mu}_F) = \widehat{\mu}_D - \widehat{\mu}_N$ holds by cancellation for every assignment and outcome realization.

For the pairwise null $p = q$, condition on the model assigned nuisance policy r. The original uniform bijection leaves a fair coin over the two assignments of p and q to the remaining model blocks. Their two observed outcomes are fixed under the pairwise sharp null, so swapping the focal labels changes only the sign of $d_{j,pq}$ and gives equation (15). Assignments are independent across prospectively written triplets, which yields the convolution in equation (16). This conditioning is essential: permuting all three labels instead would test the stronger global sharp null and need not be valid when r changes outcomes.

For the confidence statement,

$$D_{j,pq} = \sum_m \mathbf{1}\{Z_{jm} = p\}Y_{jm}^{obs} - \sum_m \mathbf{1}\{Z_{jm} = q\}Y_{jm}^{obs}$$

lies in $[-1, 1]$ and has conditional expectation $\frac{1}{3}\sum_m [Y_{jm}(p) - Y_{jm}(q)]$. The $D_{j,pq}$ are independent across triplets, though not identically distributed. Hoeffding's bounded-variable inequality therefore gives

$$\mathbb{P}\left(\left|J^{-1}\sum_j D_{j,pq} - \tau_{pq}\right| > h\right) \le 2\exp(-Jh^2/2).$$

Substituting equation (17) makes this probability at most α/K; a union bound over the K declared contrasts proves simultaneous coverage. Position-specific outcome levels do not affect either argument: positions are fixed units and policies are randomized over them. Treatment- dependent spillovers are different. They make Y_{jm} a function of the full assignment vector and invalidate the direct-effect interpretation above; the released position panel can reveal some resulting heterogeneity but cannot prove the assumption. □

Corollary 5 (Randomized position-zero carryover sensitivity)**.** *Let M_{j0} be the model block scheduled first in triplet j, let A_{j0} be its assigned first policy, and set $Y_j^0(p) = Y_{j,M_{j0}}(p)$. Model order is fixed before the independent policy shuffle. Conditional on the written triplets, realized $(M_{j0})_{j=1}^J$, and position-zero policy counts $n_p^0 > 0$, define*

$$\mu_p^0 = \frac{1}{J}\sum_{j=1}^J Y_j^0(p), \qquad \widehat{\mu}_p^0 = \frac{1}{n_p^0}\sum_{j:A_{j0}=p} Y_j^{obs}. \tag{18}$$

Then $\widehat{\mu}_p^0$ is design-unbiased for μ_p^0. For any pair p, q, conditioning additionally on the nuisance-policy membership gives the exact two-arm hypergeometric Fisher law. Moreover, with

$$h_p^0 = \sqrt{\left(1 - \frac{n_p^0 - 1}{J}\right)\frac{\log(6/\alpha)}{2n_p^0}},$$

the three policy-mean intervals $[\widehat{\mu}_p^0 \pm h_p^0] \cap [0, 1]$ and all contrasts propagated from them have simultaneous coverage at least $1 - \alpha$. The position-zero outcome cannot be affected by an earlier H95 model block in the same triplet.

Proof. Conditional on the realized first model blocks and the count vector, the policy labels at position zero are uniform over all assignments with counts $(n_p^0)_{p\in\mathcal{P}}$. Each arm is therefore a simple random sample without replacement from the J fixed values $(Y_j^0(p))_{j=1}^J$, proving unbiasedness. For a pair p, q, condition on which triplets received the nuisance third policy. Under the pairwise sharp null, the pooled focal outcomes are fixed and the positive-arm success count is hypergeometric with the realized focal counts. Finally, the sampling-without-replacement Hoeffding–Serfling bound gives failure probability at most $\alpha/3$ for each policy mean at radius h_p^0; a union bound and the triangle inequality prove the simultaneous mean and contrast statements [Serfling, 1974]. By construction, triplet position zero precedes positions one and two. The result does not rule out interference from unrelated traffic before the triplet. $\square$

The primary protocol outcome treats an absent planned first request, a mismatched first policy, a duplicate first record, or an auditable provider-control failure as zero. This defines an intent-to-treat deliverability outcome rather than a per-protocol completion rate. By contrast, an unknown or malformed value on an otherwise valid recorded request is measurement missing. Because the Fisher law requires the two focal observed outcomes in every triplet, one unknown value prevents complete-data randomization inference for every contrast containing that policy; the implementation conservatively suppresses all point estimates and reports $[0, 1]$ bounds. The paired Student-t intervals treat the realized triplets as a sample and are descriptive companions, not inversions of equation (16). The wider Hoeffding intervals report what boundedness and the randomized triplet design alone guarantee [Hoeffding, 1963].

The prerelease validation enumerates all $2^2 = 4$ conditional assignments of a two-triplet fixture and agrees with the pairwise convolution tails to machine precision. A 100,000-draw conditional-swap simulation remains only as a discrepancy audit; exact support mass must equal one within 10^{-12}, and a maximum tail discrepancy above 0.01 stops release. In two mixed-null schedules with 5,000 randomized experiments each, the corrected test's worst elementary and Holm true-null rejection is 4.06%; the superseded all-policy law reaches 8.12%. Across five fixed schedules, maximum absolute estimator bias is 0.00167, worst paired-t family coverage is 95.52%, and worst design-Hoeffding family coverage is 99.90%. The last guarantee comes from the inequality, not the simulation. Mean interval widths are 0.290 and 0.540, respectively.

Separate adversaries replace a valid outcome with `unknown`, delete and mismatch planned first requests, corrupt a provider-control length, concentrate every triplet in one six-hour bin, and verify that leave-one-model-out analysis drops whole triplets. The full repository suite has 563 passes. These checks validate the release code and claim boundary. They are not H95 outcome evidence: the fixed horizon remains five of 120 at the pinned outcome-free revision.

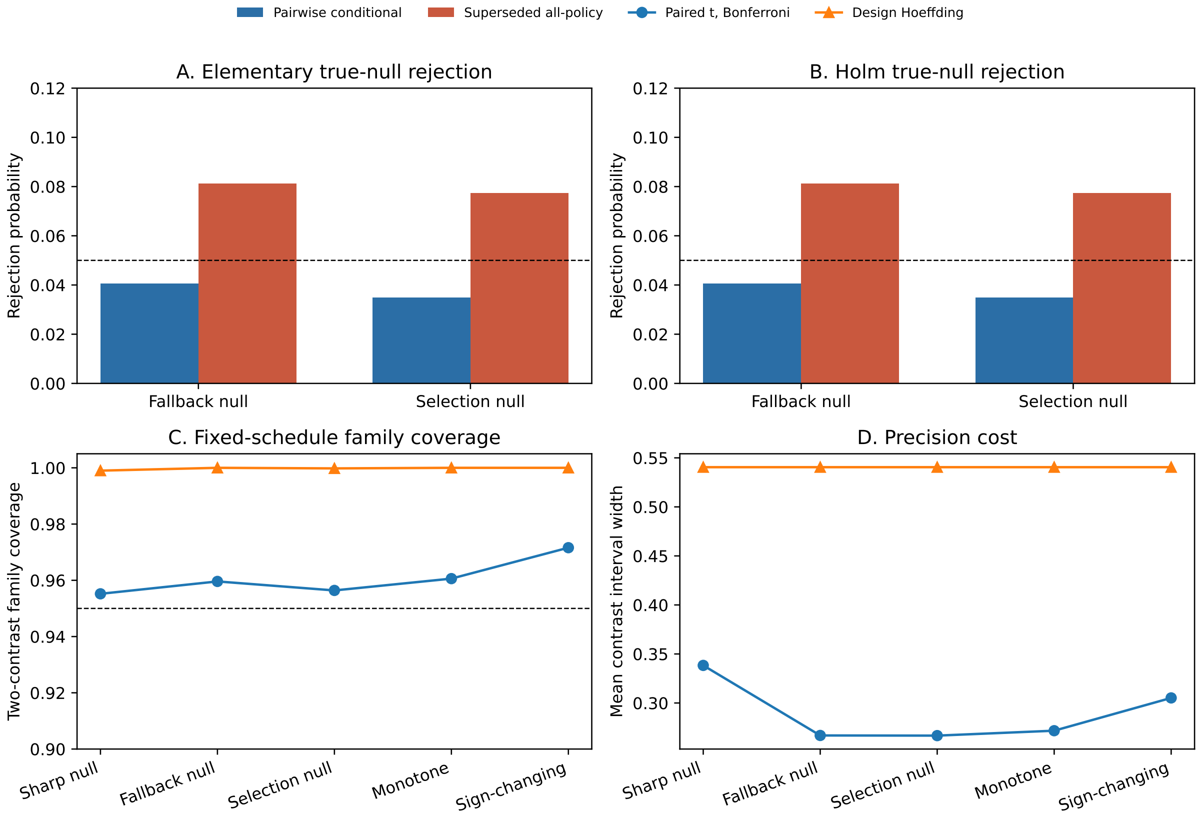

Figure 5: Outcome-blind H95 validation over fixed binary potential-outcome schedules. Panels A–B hold one elementary pair at its sharp null while the nuisance policy has an effect; the dashed line is 5%. Panels C–D compare the descriptive Bonferroni paired-t family with the simultaneous bounded-outcome design interval. These are assignment and reporting audits, not H95 outcomes.

The position-zero audit plants treatment-dependent carryover that can improve a later block after delegated policy appeared earlier while holding every direct policy effect at zero. Across 5,000 120-triplet assignments at five spillover strengths, the primary three-block hidden-selection estimator reaches absolute bias 0.2437. The position-zero estimator remains within 0.00103 of zero, with 100% observed design-family coverage, but its mean interval width is 0.810. Thus the sensitivity removes the planted channel by discarding two-thirds of the blocks; it is not a more precise replacement. Measurement missingness on a position-zero row suppresses its point and test output, while missingness only on later rows does not erase the separately identified first-block sensitivity.

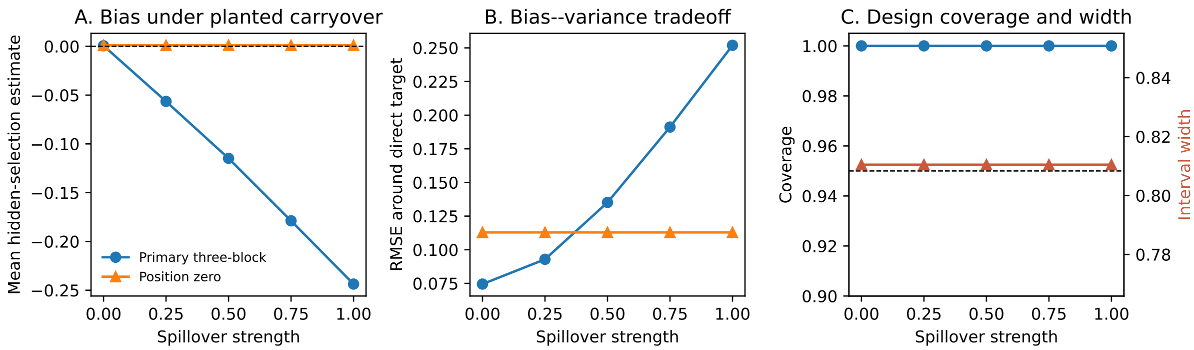

Figure 6: Outcome-blind planted-interference validation. Delegated policy can change later H95 blocks but never the first block. Panel A shows resulting bias, Panel B the bias–variance tradeoff, and Panel C the conservative position-zero design coverage and width. The construction demonstrates what the sensitivity detects; it is not evidence that actual carryover exists.

E Companion free-entry benchmark

This benchmark is retained to delimit a possible reliability/business-stealing wedge. It is not calibrated or validated by H81.

Proposition 6 (Finite entry and the reliability/business-stealing wedge)**.** *An epoch contains $D > 0$ jobs. Each of n symmetric providers is independently deliverable with probability $a \in (0, 1)$, has no binding capacity conditional on delivery, incurs marginal cost c, and pays setup cost $F > 0$. A successful job has social value $v > c$ and is paid a platform-set price p; the router shares completed jobs symmetrically among deliverable providers. Let $S_n = 1 - (1 - a)^n$ and $S_0 = 0$.*

The welfare function and a provider's expected profit are

$$W_n = D(v - c)S_n - nF, \qquad \pi_n = (p - c)D\frac{S_n}{n} - F. \tag{19}$$

The gross social value of the nth entrant is $R_n^W = D(v - c)a(1 - a)^{n-1}$, which is strictly decreasing, so a greatest welfare-maximizing count n^W is finite. With $n^E = 0$ when no provider can break even, the greatest symmetric free-entry count is also finite and, for $p > c$, satisfies

$$n^E \leq \left\lfloor \frac{(p - c)D}{F} \right\rfloor. \tag{20}$$

The entrant's gross private return is $R_n^E = (p - c)DS_n/n$. At $n \geq 2$, if $p - c = v - c$, then $R_n^E > R_n^W$ and hence $n^E \geq n^W$. More generally, there is underentry when $R_n^E < F < R_n^W$ and overentry when $R_n^W < F < R_n^E$.

Proof. Since $S_n - S_{n-1} = a(1 - a)^{n-1}$, $W_n - W_{n-1} = R_n^W - F$ decreases strictly in n and is eventually negative. Free entry requires $(p - c)DS_n/n \geq F$; because $S_n \leq 1$, this implies equation (20). Finally, $S_n/n = (a/n)\sum_{k=0}^{n-1}(1 - a)^k > a(1 - a)^{n-1}$ for $n \geq 2$. Comparing each gross return with F yields the two directional wedges. □

The proposition adapts the business-stealing logic of Mankiw and Whinston [1986] to reliability created by router substitution. It is partial equilibrium conditional on (D, p, a, c, F), not a first welfare theorem for the observed market. Public quote and owned-probe data do not identify any of those primitives jointly.

We validated the companion analytic results independently of the market data. Across 20 detection-threshold cells, simulated residuals agree with the analytic identity within 2.23 Monte Carlo standard errors and have the correct sign away from the threshold. The revenue-share coefficient identity holds to 10^{-14}, including its heteroskedasticity-robust standard error. All 144 free-entry grid cells satisfy the analytic ceiling and the equal-margin overentry result without hitting the search boundary. Finally, 20,000 random coarsening constructions produce 60,198 leader-change witnesses and no construction failure. These checks detect algebra and implementation errors; they do not calibrate the model to the observed marketplace.

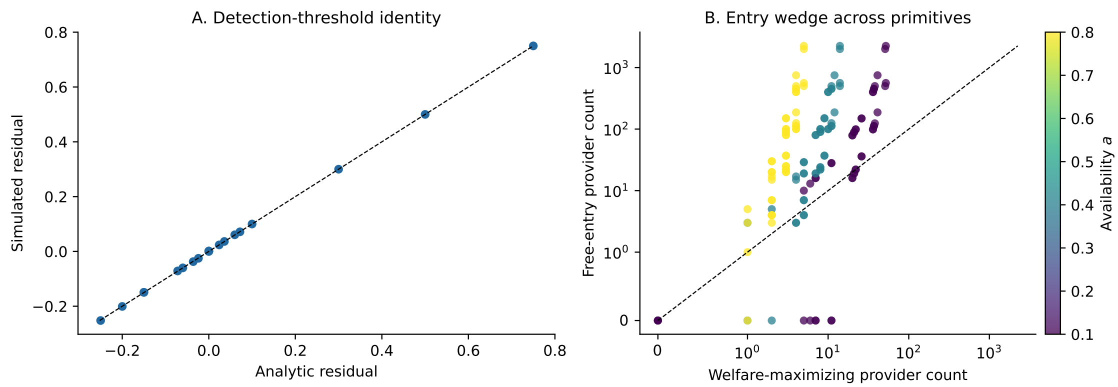

Figure 7: Logical and numerical validation of companion theory. Panel A plots the simulated residual from the menu-detection experiment against its analytic counterpart; the dashed line is equality. Panel B compares the integer welfare-maximizing and zero-profit provider counts across the primitive grid on symmetric-log axes; color records provider availability and the dashed line is equality. Dispersion around equality is the model's entry wedge, not an empirical estimate of the number of providers.

F Secondary-result detail

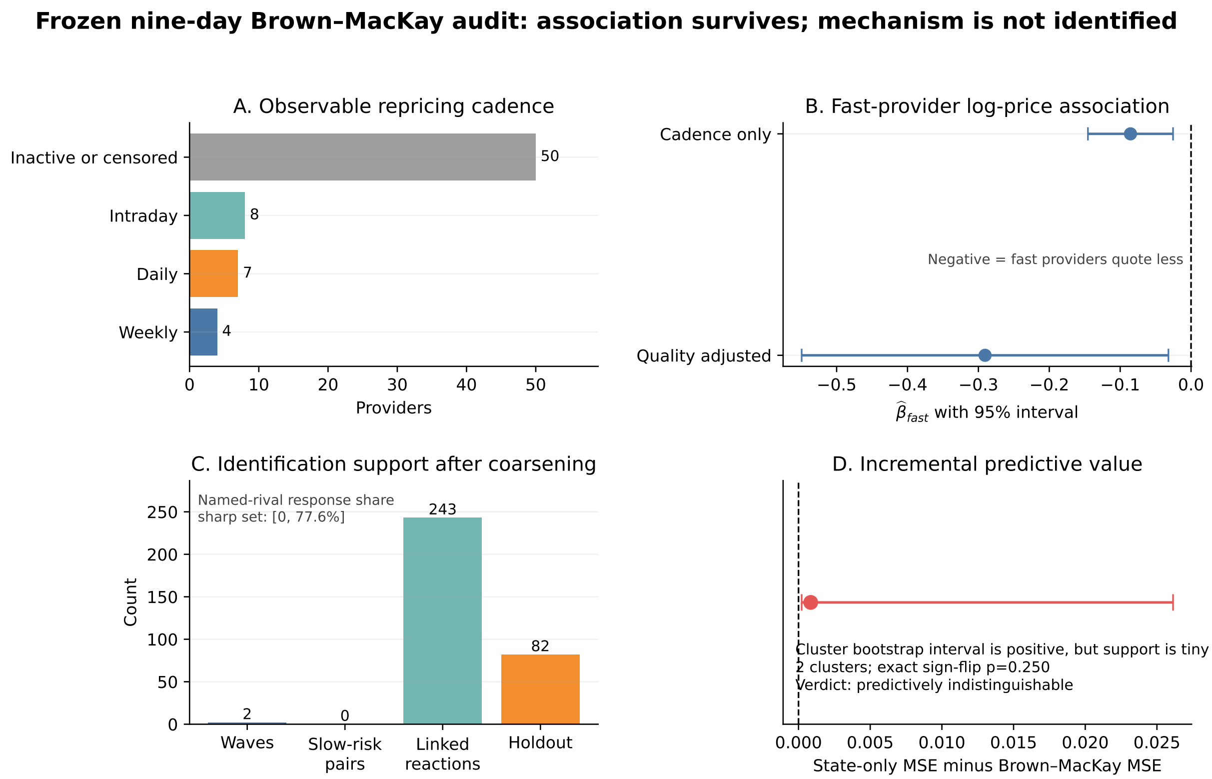

Figure 8: Corrected frozen Brown–MacKay audit. Negative cadence coefficients survive public-quality adjustment, while two frozen waves, zero slow-initiator response comparisons, and only two predictive clusters do not identify the mechanism. The 243 uniquely ordered candidates among 313 events imply the sharp named-rival response-share set $[0, 77.6\%]$.

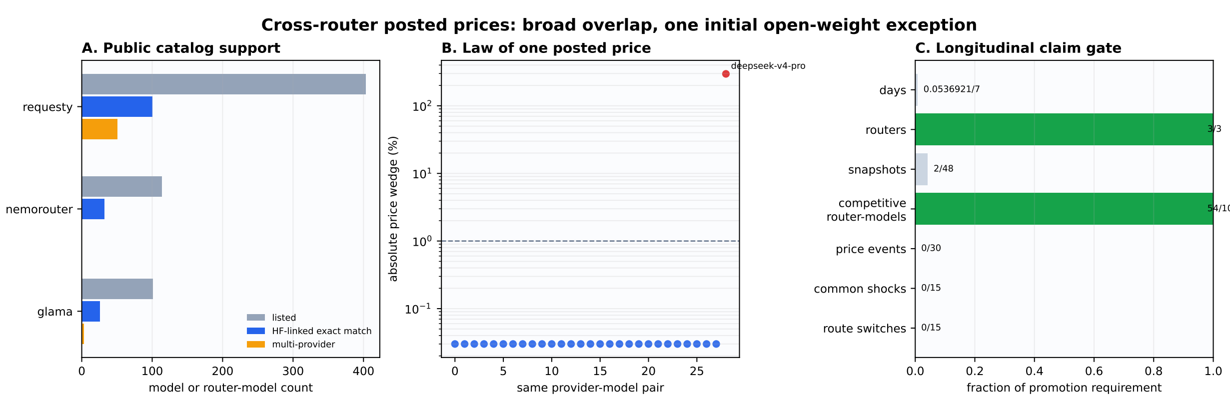

Figure 9: Discovery cross-router posted-price audit. Among 29 Hugging-Face-linked exact provider–model pairs, 28 have identical component prices. The DeepSeek exception is cross-sectional; the right panel records the longitudinal gates. H94 excludes this entire discovery cross-section and uses only snapshots after its 2026-07-17 04:30:20 UTC activation; one eligible snapshot per router is currently present and all event gates remain closed.

Fixed effects	Published $\widehat{\beta}$ (SE)	Current $\widehat{\beta}$ (SE)	Difference p
Date	-0.48 (0.15)	-0.352 (0.115)	0.498
Date and model	-1.07 (0.23)	-1.575 (0.281)	0.164
Date×model and provider×model	-1.02 (0.13)	-0.020 (0.409)	0.0197

Table 10: Published and current open-weight prompt-price elasticity ladder. The difference test treats the samples as independent; source construction and calendar windows are not identical.

G Revenue-stationarity and bounded-gap protocol

For provider i in model–time market g, let $q_{ig} > 0$ be tokens, $p_{ig} > 0$ price, $s_{ig} = q_{ig}/Q_g$ quantity share, and $e_{ig} = p_{ig}q_{ig}/R_g$ public revenue-proxy share, where $Q_g = \sum_j q_{jg}$ and $R_g = \sum_j p_{jg}q_{jg}$.

Proposition 7 (Quantity-share and revenue-share equivalence)**.** *For any common sample, positive WLS weights, market fixed effects, and additional controls, let $\widehat{\beta}_s$ and $\widehat{\beta}_e$ be the coefficients on $\log p_{ig}$ in otherwise identical regressions of $\log s_{ig}$ and $\log e_{ig}$. Then*

$$\widehat{\beta}_s = \widehat{\beta}_e - 1. \tag{21}$$

The fitted residuals are identical, so classical, heteroskedasticity-robust, and cluster-robust standard errors for the price coefficient are identical. The same result holds after exact multiway fixed-effect absorption.

The accounting relation is

$$\log s_{ig} = \log e_{ig} - \log p_{ig} + \log(R_g/Q_g). \tag{22}$$

Proof of proposition 7. Equation (22) holds observation by observation. Let M be the weighted residual-maker for the common market effects and controls, and write $x = \log p$, $y_s = \log s$, and $y_e = \log e$. Because $\log(R_g/Q_g)$ lies in the market-effect span, $My_s = My_e - Mx$. Frisch–Waugh–Lovell gives

$$\widehat{\beta}_s = (x'WMx)^{-1}x'WMy_s = \widehat{\beta}_e - 1.$$

Moreover, $My_s - \widehat{\beta}_s Mx = My_e - \widehat{\beta}_e Mx$, so the regression residuals are identical. The transformed regressor is also common, which makes every residual-based classical, heteroskedasticity-robust, or cluster-robust covariance estimate for the price coefficient identical. Exact multiway absorption simply constructs the corresponding M. □

For provider i, define the public gross-revenue proxy $\widetilde{R}_i = p_i s_i Q$, where p_i is the listed token price, s_i is its provider share within a model-day, and Q is aggregate model-day tokens. If the provider's effect on Q is negligible and marginal cost is excluded, an interior gross-revenue optimum requires

$$1 + \frac{\partial \log s_i}{\partial \log p_i} = 0. \tag{23}$$

A cross-sectional coefficient near minus one is not this derivative; by proposition 7, it is exactly a zero price coefficient for public revenue-proxy share under the same design. The prospective H91 protocol therefore decomposes log price into provider–model means and within-provider/model deviations under model-day fixed effects. Its closest published replication uses open-weight models, listed prompt prices, and provider-clustered covariance; first-difference price events add one-day pretrend and post-event falsifications.

The preregistered equivalence region is $[-1.25, -0.75]$. Promotion requires 30 panel days, 200 listed-price changes, at least 25% of provider–model entities changing price, at least 2% within-entity log-price variance, and equivalence in both the effective-price within estimate and listed-price event estimate. Until all gates pass, the cross-sectional equality is reported only as a revenue-stationarity coincidence.

A separate WCV6 sensitivity extends an estimated local share elasticity η into a bounded logit counterfactual. For observed share s and price multiple m, it sets the latent-weight elasticity to $\eta/(1-s)$ so that $\partial \log s(m)/\partial \log m = \eta$ at $m = 1$, lets total demand respond to the share-weighted price index, and searches $m \in [0.25, 4]$. Its gap $1 - \widetilde{R}_i(1)/\max_m \widetilde{R}_i(m)$ is conditional on that functional form and elasticity. It is neither observed profit nor a simultaneous equilibrium loss; a boundary optimum leaves the global distance unidentified.

H Proofs of the observational-equivalence results

Proposition 8 (Coarsened-timing observational equivalence)**.** *Let latent provider quotes be right-continuous paths observed only at grid times $t_k = k\Delta$. Consider an interval in which at least two providers change between t_{k-1} and t_k. If an administered-menu model permits provider-specific scheduled updates driven by latent public state, and a reactive model permits a provider to update within less than Δ after observing a rival move, then the sampled transition is compatible with both models. Moreover, every changed provider can be made the latent first mover in a reactive path that produces the same sampled quotes. Hence neither within-bin leader identity nor rival-triggered response is point identified from the snapshot panel.*

Corollary 9 (Sharp named-rival response bounds)**.** *For each of N sampled price events, let $U_e = 1$ when the event has a unique most-recent strictly prior rival move inside a declared lookback window, and let $R_e = 1$ when that named move caused the event. The sample response share $\theta = N^{-1} \sum_e U_e R_e$ satisfies*

$$0 \le \theta \le \frac{1}{N} \sum_e U_e. \tag{24}$$

Without restrictions beyond proposition 8, both endpoints are attainable.

Proof of proposition 2. For the nonreactive construction, let the public menu contain the finite union of price levels observed for the model. At each observed provider change, schedule that provider's private refresh clock between the two snapshots and let its state-dependent selector choose the observed terminal menu point. Providers that do not change retain their previous selection. The selector depends on the public menu state and provider state, not on a rival quote, yet reproduces the entire sampled panel. In particular, choosing a menu point already occupied by a rival creates both a cross-sectional match and a lagged exact landing without strategic response.

For the reactive construction, retain the same sampled terminal prices. For any chosen subset of the L exact-landing events, define the mover's response map at the realized history to select the matched rival quote; use the scheduled rule for all other events. This produces the same observations while classifying any integer number from zero through L as exact strategic landings. Dividing by N attains every grid point in $[0, L/N]$; mixtures attain the full interval. No event outside the L exact landings can enter the defined exact-landing estimand, proving sharpness. □

Proof of proposition 3. Without reactive replacement, $\Pr(Y_e = 1) = p_e$. A replacement forces $Y_e = 1$; otherwise the nonreactive draw remains. Hence $\Pr(Y_e = 1) = \rho + (1-\rho)p_e = p_e + \rho(1-p_e)$. Subtracting q_e gives equation (12). If $q_e \ge p_e$ and $p_e < 1$, solving the strict inequality $p_e - q_e +$

$\rho(1-p_e) > 0$ gives the stated threshold. Linearity of expectation gives the event-mean version when ρ is common and the event panel is fixed. □

Proof of proposition 8. Fix an observed transition and let S be the providers whose sampled quotes change. Choose $\epsilon > 0$ with $(|S|+1)\epsilon < \Delta$. For a nonreactive path, order the providers in S arbitrarily and let provider i replace its old quote with its observed terminal quote at time $t_{k-1} + r_i\epsilon$. Declare these provider-specific update times and terminal quotes to be functions of the latent public state at the start of the interval. No rule uses a rival quote, and the sampled endpoints match the data.

For a reactive path, choose any provider $j \in S$ and any ordering of S with j first. Let j move at $t_{k-1} + \epsilon$. After the rth move, let the next provider observe that move and update at $t_{k-1} + (r+1)\epsilon$; define its response map at this realized history to equal its sampled terminal quote. Every delay is strictly positive and less than Δ, and the terminal vector again matches the data. Repeating the construction with another j reverses the latent leader without changing either sampled endpoint. Providers outside S remain constant. Applying the construction interval by interval establishes the result for the full sampled panel. □

Proof of Corollary 9. An event with $U_e = 0$ cannot contribute to the named-rival estimand, and $R_e \in \{0,1\}$, which gives equation (24). The lower endpoint is attained by the scheduled construction in proposition 8. For the upper endpoint, use the reactive construction for every $U_e = 1$ event, assigning its unique strictly prior rival as the trigger; unrestricted response maps preserve all sampled terminal quotes. Thus neither endpoint can be tightened from the snapshot panel alone. □

The proposition is intentionally nonparametric. It does not say scheduled and reactive models remain equivalent after imposing a known common-state process, a parametric reaction map, a lower bound on response latency, or continuous update timestamps. It says those restrictions are additional identifying data or assumptions; they are not supplied by a longer panel at the same observation frequency.

I Held-out whole-market menu protocol

The held-out analyzer resolves the earliest 30 quote dates at one immutable data revision. Dates 1–15 estimate feature means, feature scales, and one persistent landing increment; dates 16–30 are never used for fitting. Before all 30 dates exist, the code reads only the distinct-date ledger and emits no price or event table.

For event e at prior timestamp t^-, every other model observed at exactly t^- is a candidate decoy market if it has at least as many endpoints as the focal rival set. Markets are matched by Euclidean distance after training-period standardization of five pre-event features: log endpoint count, log median price, log relative interquartile range, shared-price endpoint share, and an indicator for a tied minimum. The primary retains the nearest 20 markets, requires at least five, and uses fixed 10- and 50-market sensitivities. Within decoy market j, N_j is endpoint count, H_j counts endpoints exactly at the mover's realized new price, and r_e is focal rival-set size. The conditional chance match is

$$q_{ej} = 1 - \frac{\binom{N_j - H_j}{r_e}}{\binom{N_j}{r_e}}, \qquad q_e^{\mathrm{MM}} = \frac{1}{K_e} \sum_{j=1}^{K_e} q_{ej}. \tag{25}$$

This preserves whole-market multiplicity rather than pooling endpoints.

The nonreactive benchmark is $M_0 : Y_e \sim \text{Bernoulli}(q_e^{\mathrm{MM}})$. The nested persistent-increment model is $M_1 : Y_e \sim \text{Bernoulli}(q_e^{\mathrm{MM}} + \rho(1 - q_e^{\mathrm{MM}}))$, with $\rho \in [0,1]$ fit only on dates 1–15. Promotion requires at least 100 holdout events, ten model clusters, no model above 50% of events, positive model-cluster lower bounds and leave-one-model-out estimates for both the residual $Y - q^{\mathrm{MM}}$ and holdout log-score gain, and one-sided cluster sign-flip $p \leq 0.05$ for each. Even promotion establishes only persistent same-model landing beyond this exchangeability class. The procedure conditions on the realized event and new price, so it is not a full refresh-clock or equilibrium model and cannot identify strategic response, collusion, or front-running.

J Independent grid-null algorithm

For each observed model-day (m, d) with $N_{md} \geq 2$:

1. Compute the observed median log price $\widetilde{p}_{md}$ and provider deviations $u_{imd} = \log p_{imd} - \widetilde{p}_{md}$.
2. Pool deviations over model-days after the same endpoint and variant filters used by the observed statistic.
3. Draw N_{md} deviations independently with replacement, add $\widetilde{p}_{md}$, exponentiate, and snap to the declared grid.
4. Record whether the two smallest simulated prices are equal. Aggregate over model-days and repeat the full simulation.

The primary grid is one cent per million tokens; the sensitivity is ten cents. The simulation preserves the distribution of market thickness and model-day price levels but removes cross-provider dependence. Because the deviation pool is empirical, it does not assume a parametric cost distribution. Its main limitation is that pooled deviations combine provider heterogeneity and may retain atoms from the observed coordination process.

K Randomized-crossover protocol

Each eligible block chooses a hot model with at least three publicly quoted providers. It freezes the contemporaneous provider set and computes cheapest, second-cheapest, and random pinned candidates. A deterministic pseudorandom seed permutes the four policies and the model order. The assignment row is written before requests begin. Requests use a fixed minimal prompt and one output token.

The primary outcomes are HTTP-level success and a valid generation identifier. Secondary outcomes are latency, observed spend, selected provider, and fallback attempts when exposed. Missing cost is not recoded as zero. A block is valid only if assignment replay passes, the quote snapshot predates the request, and each declared policy is attempted exactly once. The confirmatory prefix stops at 500 valid attempts per pinned arm; later rows are a separate continuation sample.

The three primary hypotheses compare default success with each pinned arm. The family uses Holm adjustment. Randomization inference permutes policy labels according to the original assignment mechanism within blocks. The first-position estimator is primary for arbitrary carryover. Pooled position-adjusted contrasts and model-policy heterogeneity are secondary. A flat cheapest/second/random gradient is reported as evidence against rank-specific stale-quote refusal, not as equivalence unless the registered equivalence margin is met.

L Steering-audit robustness

For provider–model–day cell (i, m, d), let S_{imd} be the selected-provider share among default probes, C_{imd} indicate a price cut in the prior seven days, and R_{imd} be current price rank. The audit estimates

$$S_{imd} = \alpha_{md} + \beta C_{imd} + f(R_{imd}) + \varepsilon_{imd}. \tag{26}$$

The unrestricted audit includes every provider-model-day that is currently cheapest on the public surface and has at least one default probe. Model-day fixed effects remove common demand and quote-level shocks; price-rank controls compare providers at the same current prominence. Standard errors cluster at provider and model-day in the promotion specification.

Three robustness layers are preregistered: (i) require at least one successful pinned request for the provider-model pair; (ii) use only regions and prompt lengths for which all compared providers pass eligibility; and (iii) instrument recent-cut visibility with randomized quote-surface exposure if the platform permits a safe visibility experiment. Current data support the direction in layer (i) but not its sample size. No visibility instrument has fired.

M Transparent-compute comparator

A separately preregistered Akash study links public multi-provider bid sets, the accepted lease, and exact on-chain termination reasons. The launch backfill replayed 2,929 retained closed leases with a 100% exact-ID match. Its pre-preregistration support set contains 36 linked multi-provider choices: 20 accepted prices above the lowest bid and 16 at the lowest. All 36 have the fixed 300-block follow-up, but only 20 close in that window; none records a provider, escrow, or non-owner close. This is zero-event support, not evidence of no effect.

The prospective cohort uses a second-snapshot inception rule, masks confirmatory outcomes, and has a fixed release of 2026-08-15. The comparator is excluded from the paper's promoted facts because its market object is a compute lease rather than a token-priced inference request. Its role is methodological: it shows what becomes identifiable when bids, acceptance, and termination are public.

N Claim ledger and promotion gates

Claim	Current evidence	Missing identification	Promotion gate
H81 fallback and hidden selection	84 verified blocks; counts 32, 24, 28; no outcome released	Precision and transport beyond two repeated models	Original 40-per-arm gate; terminal excluded; Holm tests, familywise intervals, and worst-case treatment/outcome bounds frozen
H80 provider-order replication	156 verified blocks; counts 37, 43, 38, 38; outcomes masked	Independent replication under different policies and support	Later 500-per-arm promotion amendment
H95 fixed-horizon replication	Five compliant triplets, 15 blocks, nine unique models; 12 legacy-unverified and three passing auditable control records; no outcome released	120 written triplets, row-level control coverage, broad audited model/time support, and cross-position robustness	Fixed 120-triplet horizon; nuisance-conditioned pairwise Fisher law, simultaneous design intervals, and lower-precision position-zero sensitivity; whole-triplet LOMO; never pooled with H81
Displayed price versus operational state	H82 fixed-price enforcement paths with failed pretrends	Exogenous enforcement timing and request-level allocation	Future-only H83 holdout
Stale-cheap pickoff channel	H84 rejects its directional prediction; backward placebo nonzero	Prospective sample and realized fill linkage	Disjoint H85 holdout
Cross-router posted-price equality	28/29 HF-linked pairs equal in one simultaneous discovery cross section, Wilson interval $[0.828, 0.994]$; 28/29 have an observed term conflict	Repeated changes, full contract comparability, eligibility, and fills	H94 has one prospective snapshot/router after 2026-07-17 04:30:20 UTC; all dynamic event gates closed
Literal front-running	Coarsened-timing bounds only	Cross-user arrival and quote-update sequence	Negotiated router/provider event logs
Global welfare loss	No identified estimate	Values, quality, costs, transfers, and capacity response	Structural measurement contract

Table 11: The ledger distinguishes a supported market fact from the stronger mechanism or welfare claim it may motivate.

O Reproducibility

All tables come from versioned Parquet and JSON artifacts carrying source and retrieval timestamps and run identifiers. Probe assignments record study and block identifiers, seed, intended policy, position, and quote-snapshot key; prompts and secrets are excluded. The repository includes analyzers, preregistrations, claim boundaries, and machine-readable summaries. The four-panel figure reads only frozen `paper_estimates.json`, not the pre-randomization legacy H80 summary. The manuscript release resolves the Hugging Face dataset head before its first query and pins every subsequent query to that immutable revision. The corrected public-input release uses immutable dataset revision `b389923ad7713bc230dd522f770aa306bf778806`. A single local snapshot of that revision ran the fixed public-only manuscript panel: 26 modules completed in the main pass, and the historical-event module completed in a supplemental pass after its public Wayback input was added to the snapshot. Prospective H80, H81, and H95 outcome analyzers are not members of this rerun list. This distinction matters because a date window alone does not prevent a late backfill from changing an earlier date. The outcome-free `manuscript_promotion_gate_summary.json` freezes the nine-day and earliest 30-day vintages and masks H80 outcomes until 500 first-position assignments per arm. The companion `manuscript_vintages` program runs PM1, PM5, and BM1–BM4 under both calendar cutoffs, canonicalizes every table, records SHA-256 hashes, and compares only the metrics fixed before the 30-day data exist. The separate `pm1_temporal_validation` program uses a completed-date 15/15 split and prior-close features; its readiness path cannot read the PM1 pricing-event table before the 30-date gate. For the focal experiments, a clean GitHub Actions runner executes after each successful data compaction. It pins one immutable input revision and queries assignment fields only while H81 or H95 is below its frozen gate. Once a gate opens, the runner first commits a timestamped access marker containing the code, input, and environment hashes; only then can the dedicated analyzer read outcome fields. A non-publishing invocation remains preflight-only even after a gate opens. The H81 release analyzer at commit `4d66fda` distinguishes binary failures from unknown outcomes, refuses complete-data inference under outcome attrition, and emits both arm-specific and intended-assignment worst-case bounds. Commit `55b5087` adds exact finite-support Fisher tails; commit `5cc0a4a` makes the retained permutation audit fail closed at a one-percentage-point discrepancy. The subsequent outcome-blind pairwise amendment holds the nuisance third arm fixed, adds a nuisance-effect size audit, and records exact minimum-count power before release. A second outcome-blind amendment adds the simultaneous finite-population interval, fixed-schedule coverage audit, and exact joint-Holm power. Commit `f170d89`, made at H95 support 4/120 and before outcome access, adds the initial finite assignment convolution, explicit structural-versus-measurement missingness, row-level provider-control coverage, paired familywise intervals, time concentration, whole-triplet leave-one-model-out, and a triplet-position diagnostic. The release now hashes every dated protocol amendment and writes redacted outcome, position, and transport audit panels. A second outcome-blind H95 proof audit at 5/120 replaces the global six-assignment law with nuisance-conditioned focal swaps, adds mixed-null size validation, and reports simultaneous bounded-outcome design intervals. A third outcome-blind H95 amendment formalizes the randomized position-zero estimator, exact secondary pairwise family, simultaneous Hoeffding–Serfling intervals, and planted-carryover bias audit without changing the primary horizon or family. The full repository test run has 551 passes at the initial H95 freeze, 557 after the H81 interval audit, 561 after the H95 pairwise- interval audit, and 563 after the H95 position-zero interference audit. The remote path publishes a content-hashed bundle exactly once and refuses automatic re-access if an orphan marker records an interrupted first run. Live preflight run `29563312069`, checking out manuscript/code head `f2fd115`, reproduced H81 counts 32/23/27 and H95 support 4/120 at their separate outcome-free revision `4fd167d674f6`, with both outcome-query flags false. The preceding data consolidation was workflow `29561825717`, whose preparation and eight shards all passed; the full public-screen rerun `29561010800` also completed analysis, memo, dataset publication, and Space publication; the independent remote-health audit `29560495633` passed. The public-input integrity repair ran remotely as workflow `29558374319`; the post-repair audit exactly reconstructed 3,219 of 3,219 completed-day pricing events and found no price or capability conflict among same-listing raw variants. The two prospective studies are never pooled. An outcome-blind preflight using `f170d89` against the same revision again found H95 support 4/120, perfect plan compliance and replay, zero missing first records, and 12 legacy-unverified provider-control rows; it did not select an outcome field. Scheduled collector run `29564165459` then checked out `f170d89` and completed successfully at 07:49 UTC. New-head compaction `29564756681` passed preparation, the 551-test suite, publication, and all eight table shards. Its automatically triggered release audit `29565268475` checked out `14ba8c6`, pinned revision `3efd953a9810`, and found H81 counts 32/24/28 and H95 support 5/120. H95 has 15/15 recorded first requests, perfect replay and plan compliance, 20% row-level control coverage, and a 100% pass rate among the three auditable records; both studies again report `outcomes_queried=false`. Exact-paper-head audit `29565662719` then checked out `d61a6b2`, reproduced the same revision and counts, and again left both outcome-query flags false. Corrected-paper audit `29566914482` checked out `244c384`, pinned revision `42334a840dc`, reproduced the same counts, and again left both outcome-query flags false. The design-interval amendment was frozen immediately after that outcome-blind provenance point. Published-paper audit `29568434722` checked out `973f900`, pinned revision `30b430e2a09`, reproduced H81 counts 32/24/28 and H95 support 5/120, and again left both outcome-query flags false. This is the provenance point for the H95 pairwise-law and design-interval amendment. Corrected-H95 audit `29569590704` then checked out `4860015`, pinned revision `f5b822812a8`, reproduced the same counts and support, and again left both outcome-query flags false. This is the provenance point for the position-zero carryover amendment.

Privacy, load, and release tiers. Capture records exclude prompts and completions. Owned experiments use a fixed minimal prompt, cap output at one token, serialize interfering studies, and bound spend. Public quote requests and router counters are rate limited; failed captures create missing intervals rather than retry storms. The public tier contains code, manifests, quote-derived tables, assignment fields, aggregated statistics, and replay seeds. Account-specific generation references and raw attempts remain restricted; released aggregates are hashed, while payloads and credentials are excluded.